

Ch 4 Classification

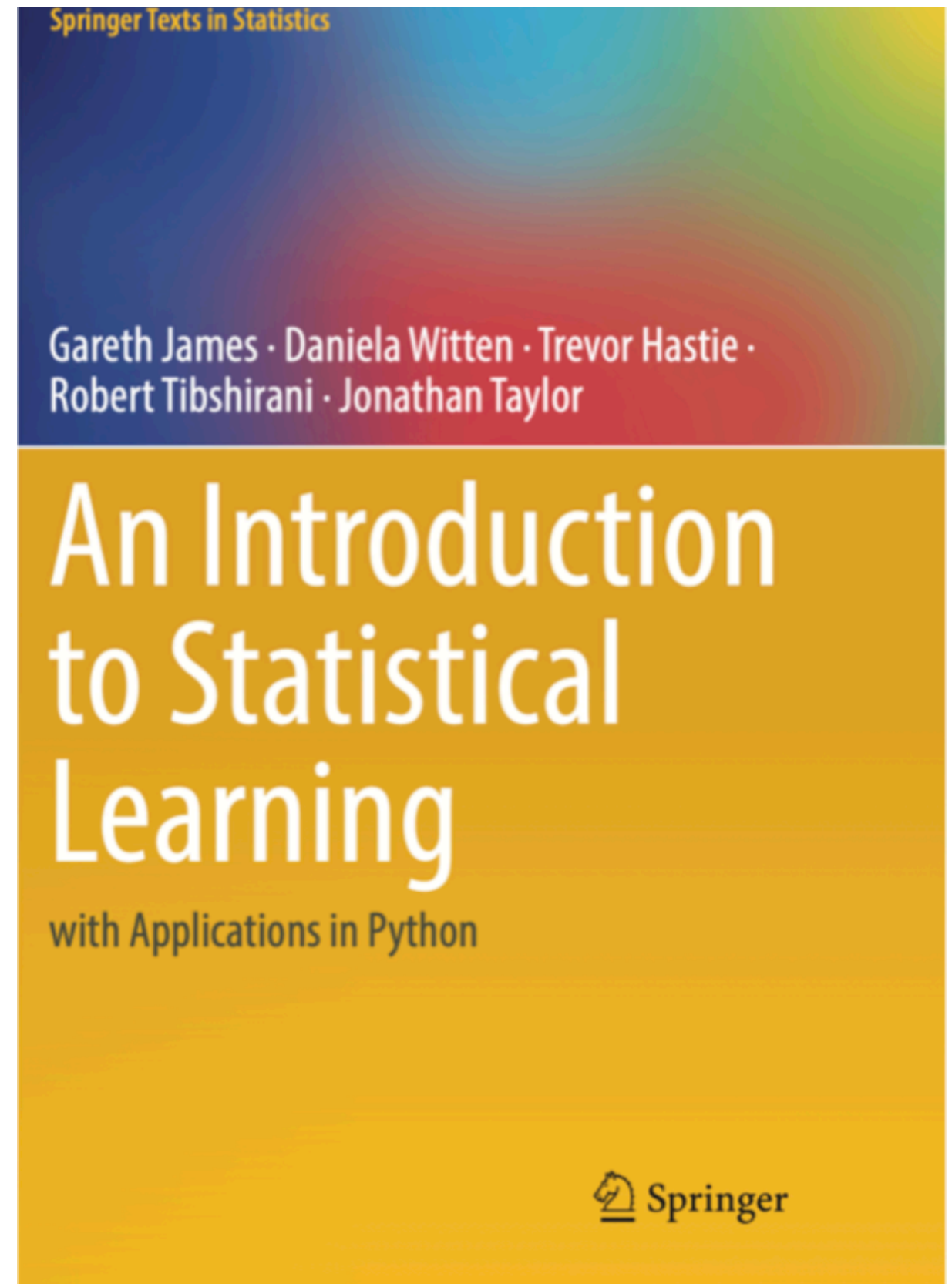
Instructor: Huei-Wen Teng

Slides modified from <https://www.statlearning.com/>

October 20, 2023

4.1

An overview of Classification



<https://www.statlearning.com/>

Q: What you did if your predictors are qualitative?

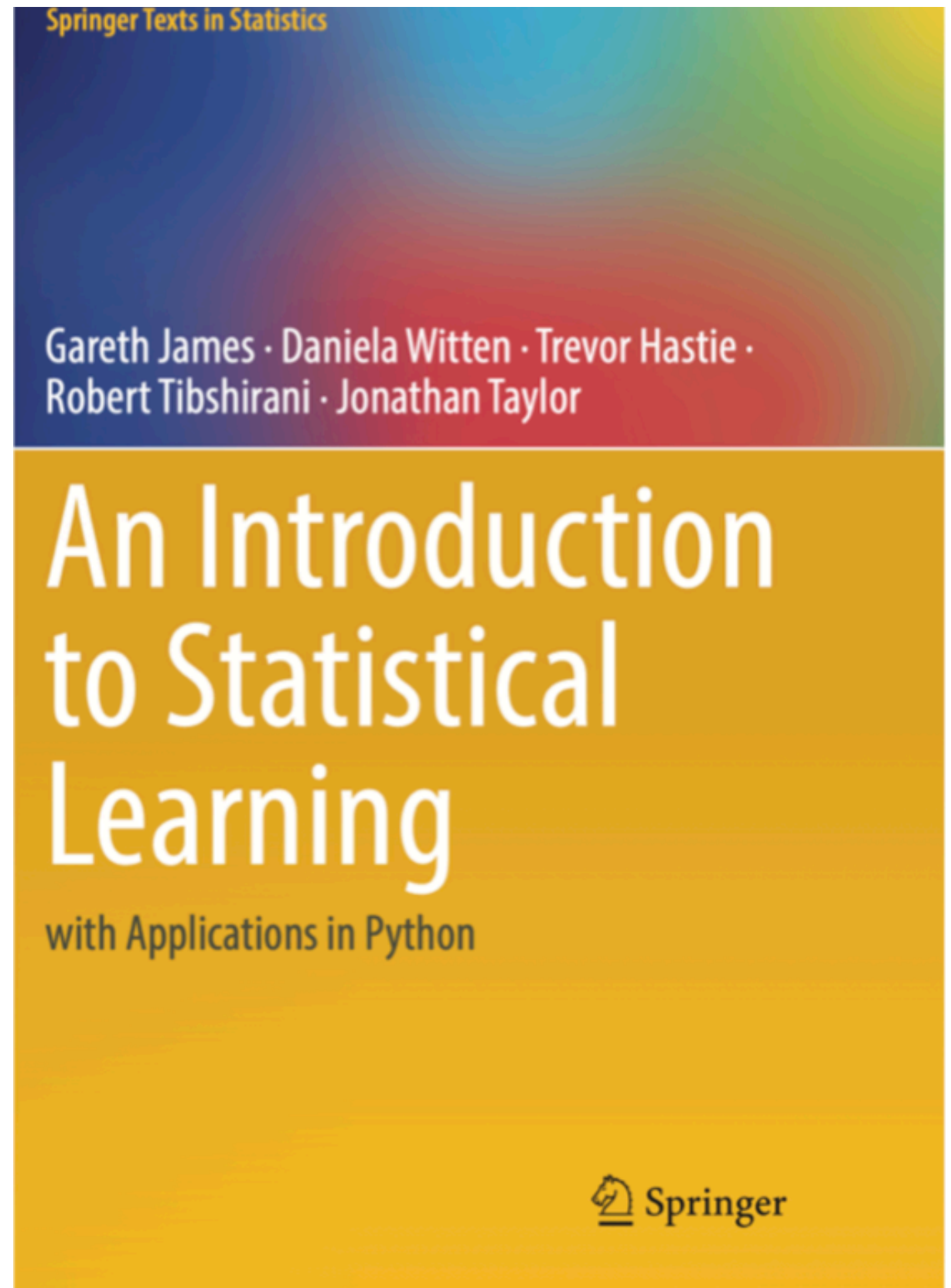
Classification

- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:
 $\text{eye color} \in \{\text{brown}, \text{blue}, \text{green}\}$
 $\text{email} \in \{\text{spam}, \text{ham}\}.$
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y ; i.e. $C(X) \in \mathcal{C}$.
- Often we are more interested in estimating the *probabilities* that X belongs to each category in $\mathcal{C}$.

For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

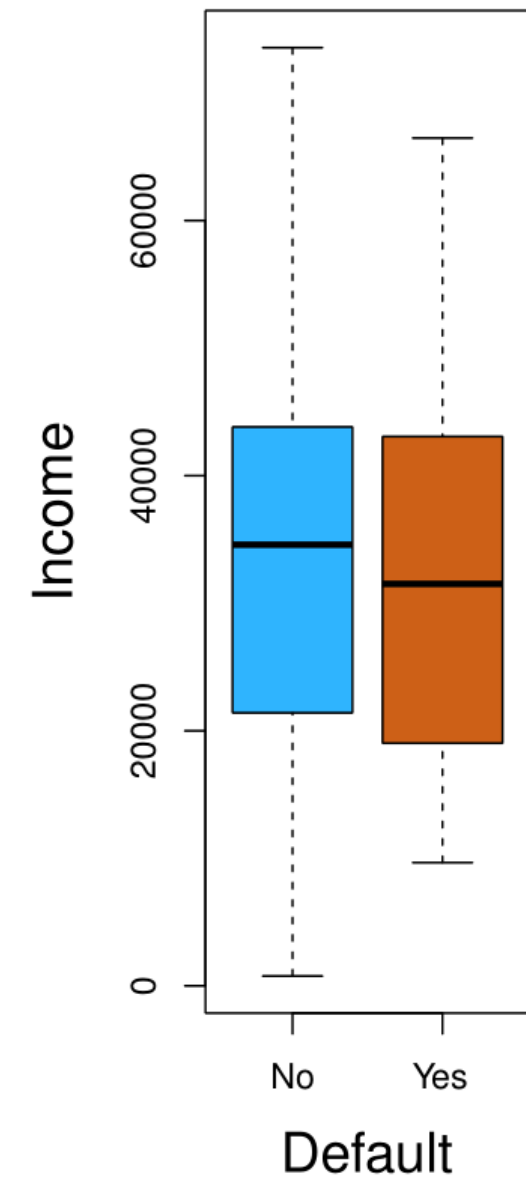
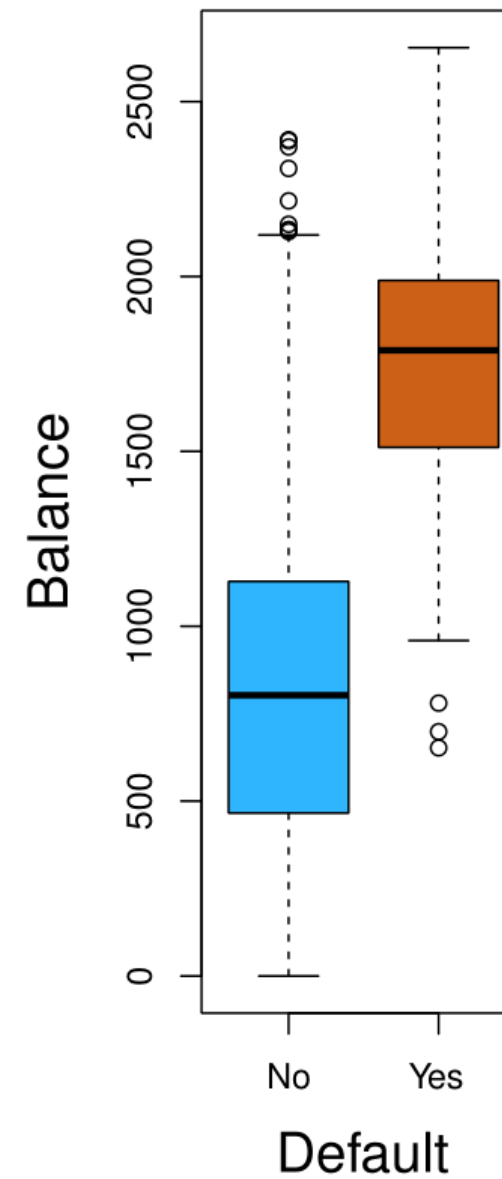
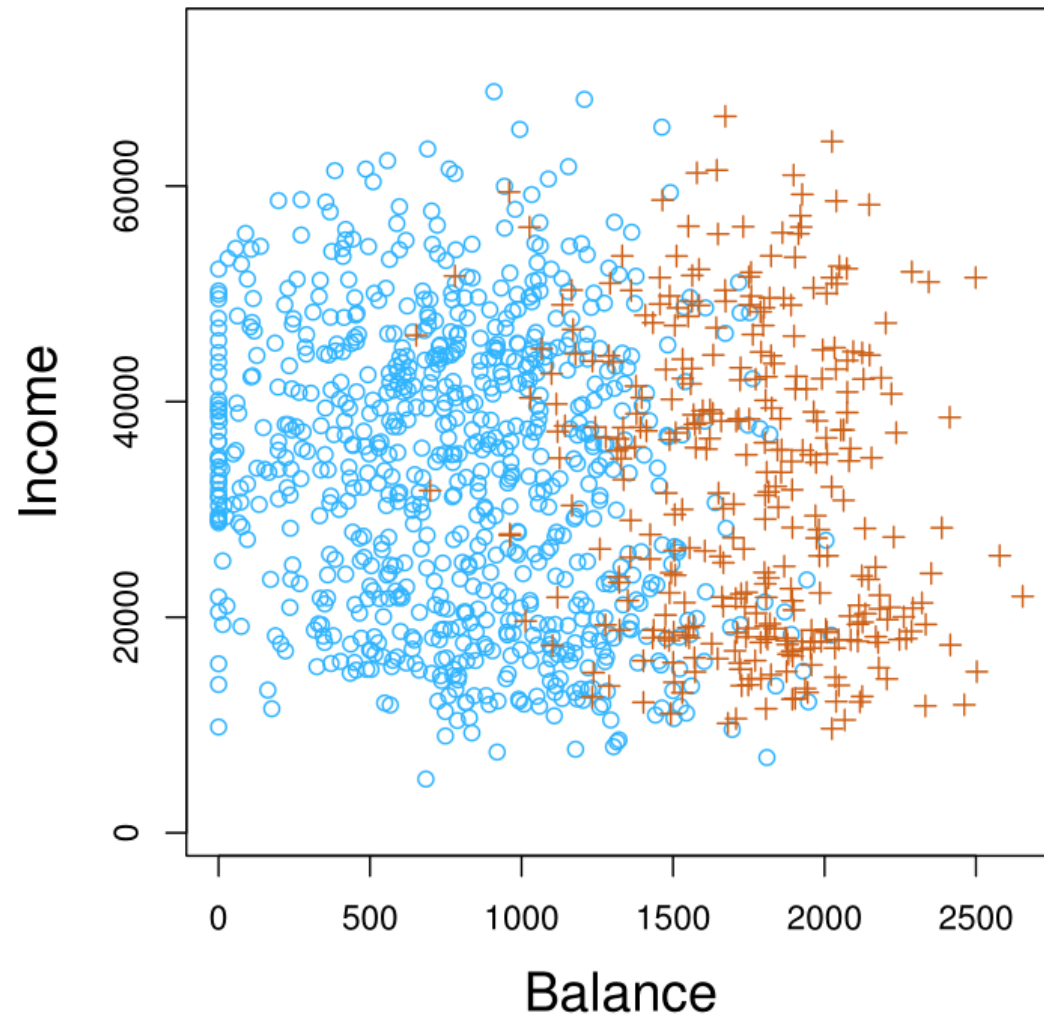
4.2

Why not linear regression?



<https://www.statlearning.com/>

Example: Credit Card Default



Can we use Linear Regression?

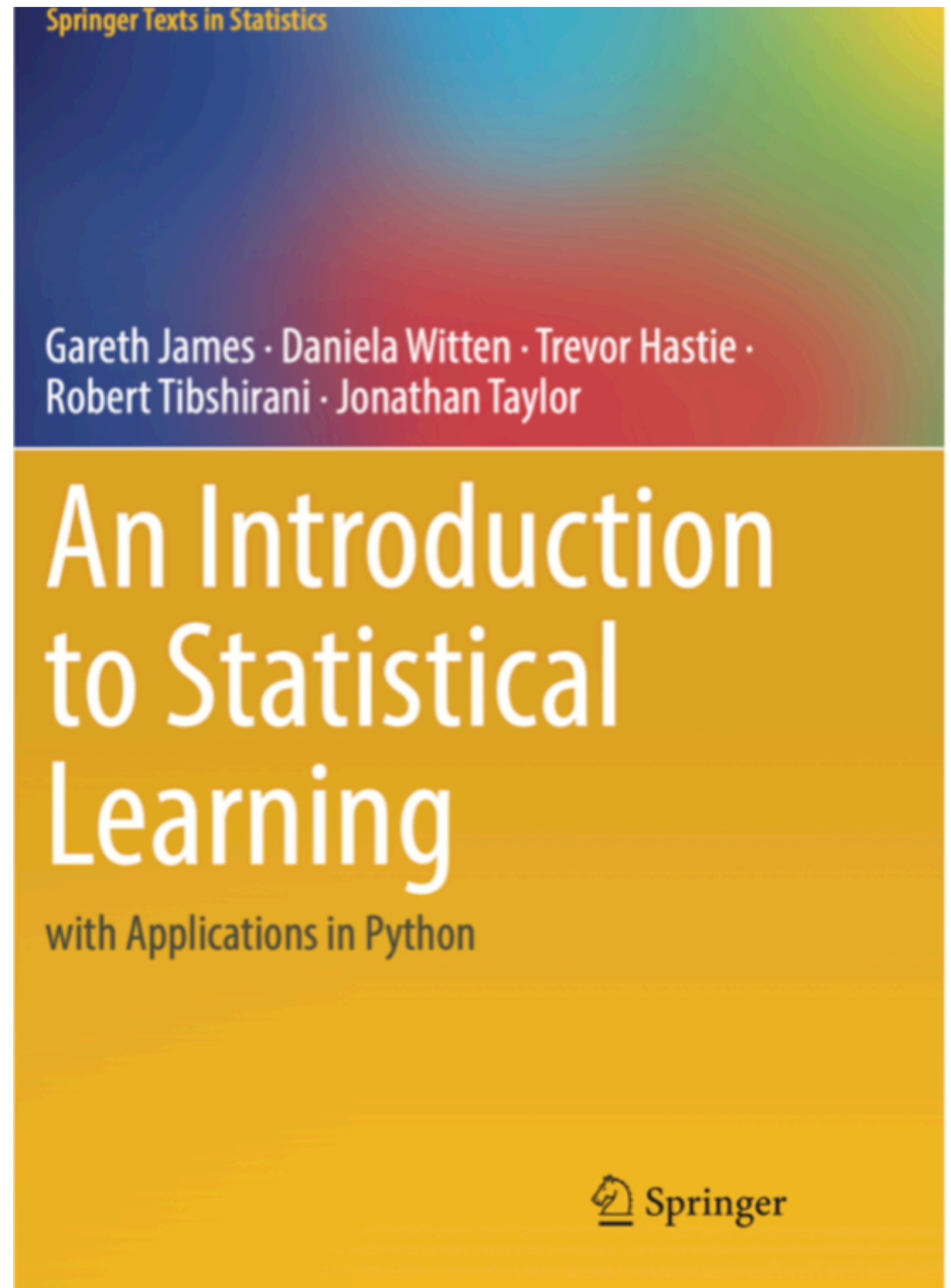
Suppose for the **Default** classification task that we code

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes.} \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as **Yes** if $\hat{Y} > 0.5$?

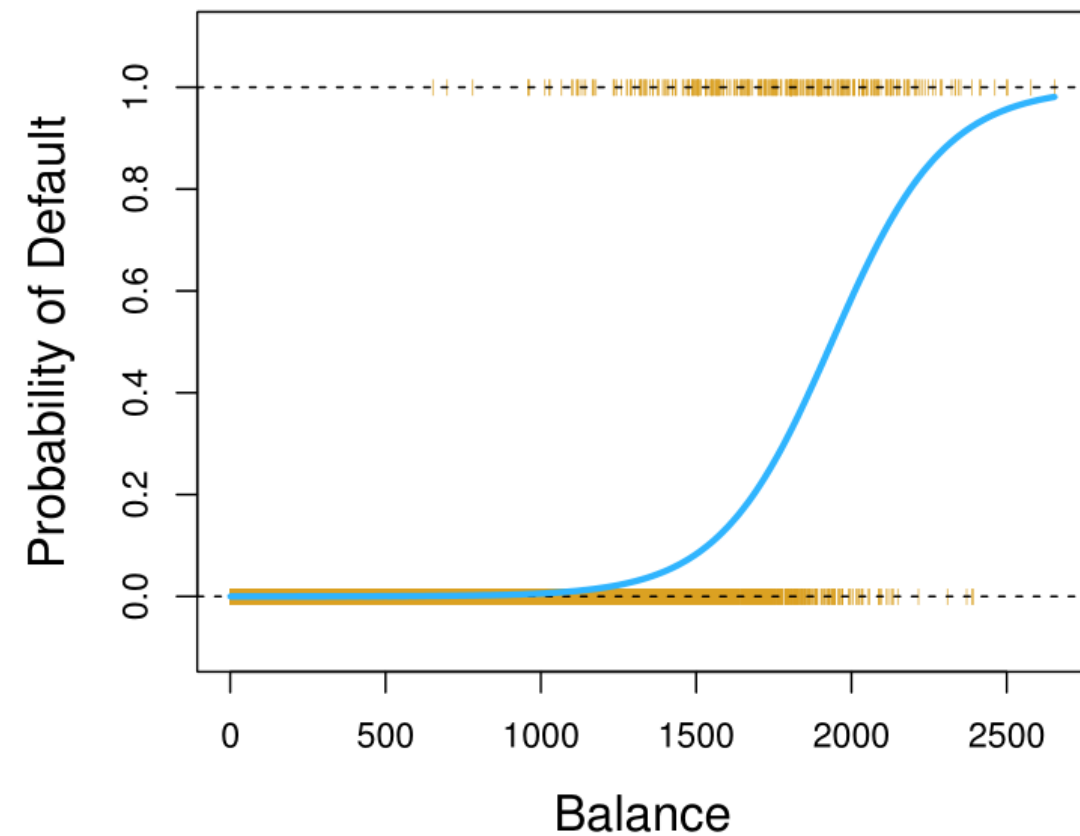
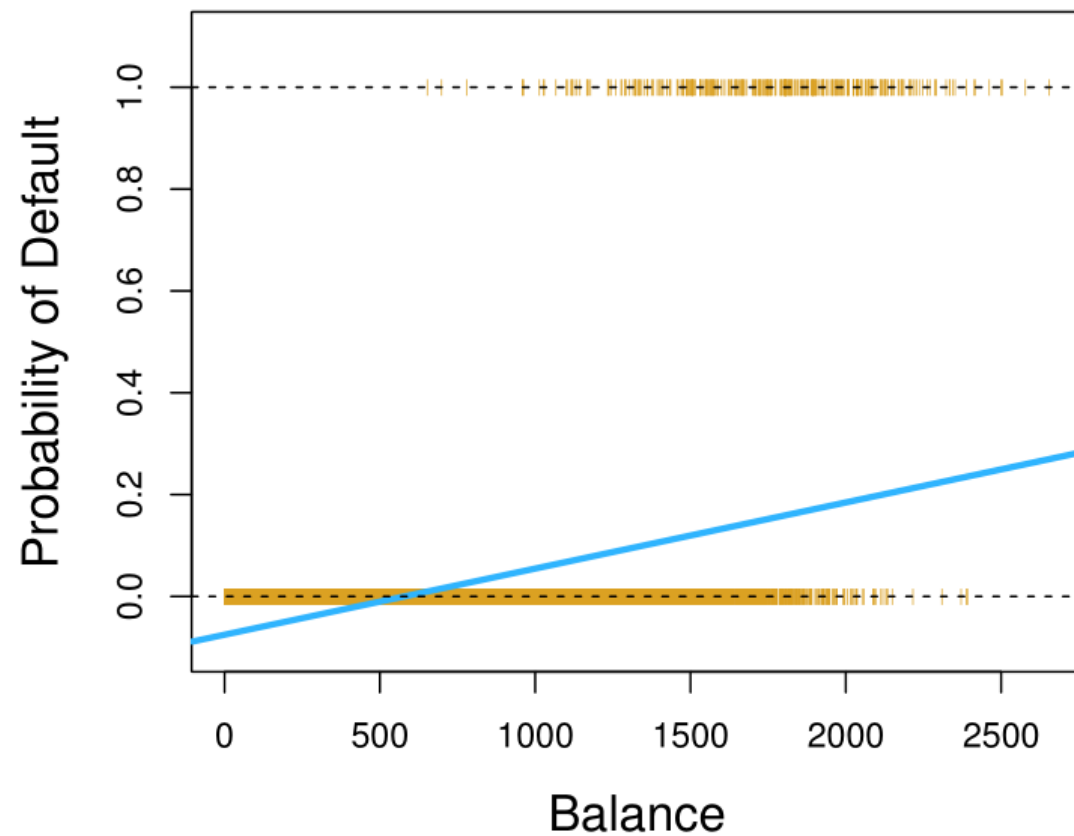
- In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to *linear discriminant analysis* which we discuss later.
- Since in the population $E(Y|X = x) = \Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.
- However, *linear* regression might produce probabilities less than zero or bigger than one. *Logistic regression* is more appropriate.

4.3 Logistic Regression



<https://www.statlearning.com/>

Linear versus Logistic Regression



The orange marks indicate the response Y , either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Linear Regression continued

Now suppose we have a response variable with three possible values. A patient presents at the emergency room, and we must classify them according to their symptoms.

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epileptic seizure.} \end{cases}$$

This coding suggests an ordering, and in fact implies that the difference between **stroke** and **drug overdose** is the same as between **drug overdose** and **epileptic seizure**.

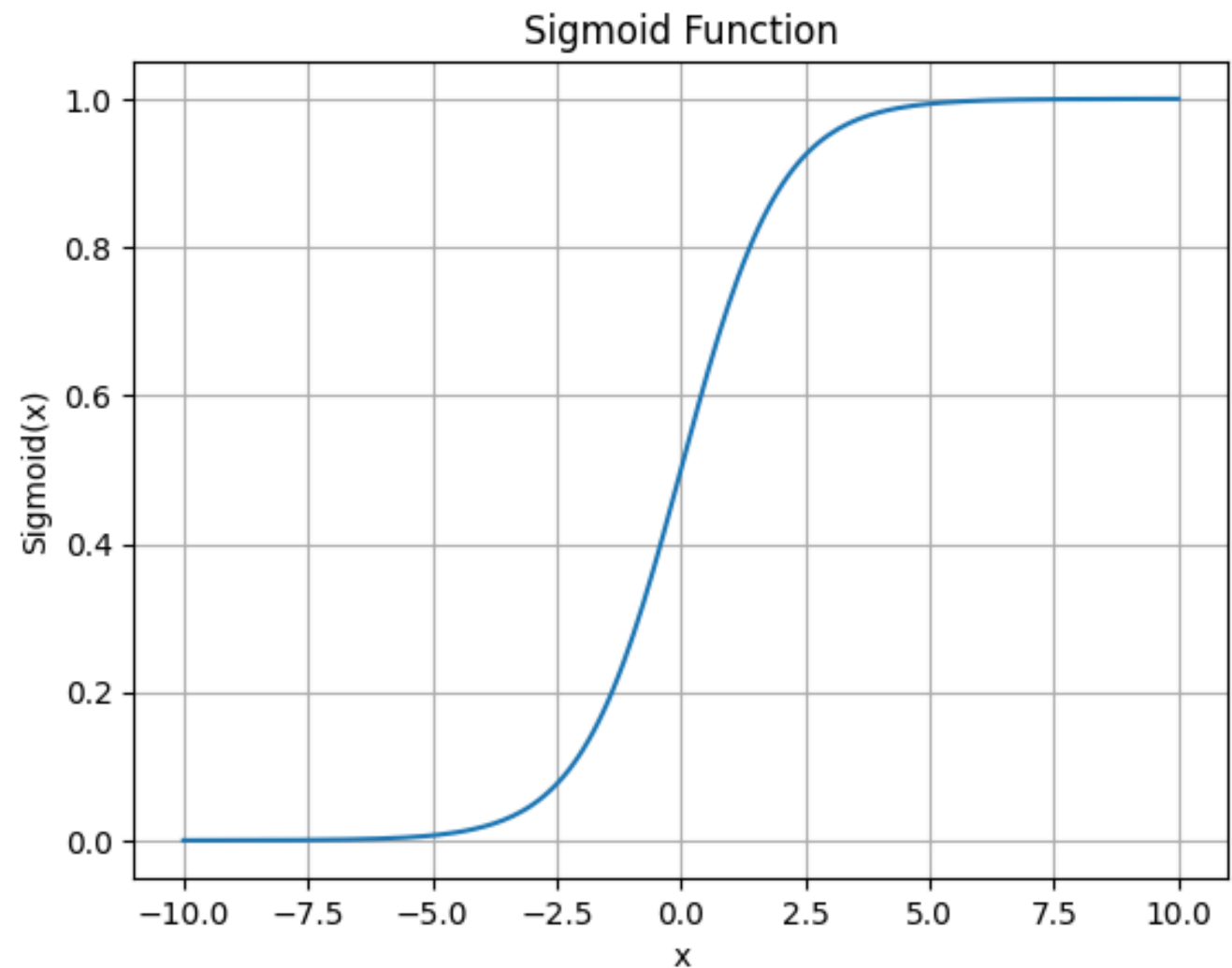
Linear regression is not appropriate here.

Multiclass Logistic Regression or *Discriminant Analysis* are more appropriate.

The sigmoid function

How do you draw it by hand?

- $\sigma(t) = \frac{1}{1 + e^{-t}}$ for
 $-\infty < t < \infty$
- Range: bounded in $[0,1]$
- Shape: S



https://colab.research.google.com/drive/1xnuBEllQrWw3lqb_dpAEuwhmD2ZZfSjd#scrollTo=DjH5fTgmaICM

Model

$(\beta_0, \dots, \beta_p)$ a vector of coefficients. Let $Bernoulli(\pi)$ denote the Bernoulli distribution with probability π . Equivalently, Eq. (1) can be expressed as

$$y|x \sim Bernoulli \left(\frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p)}} \right),$$

where the maximum likelihood estimation estimates the coefficients.

To interpret the logistic regression, we use the log-odds, which is the logarithm of the odds ratio:

Interpretations

$$\text{Log-odds} = \log \left(\frac{P(y = 1|x)}{1 - P(y = 1|x)} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p.$$

The coefficients in logistic regression offer insights into the relationship between predictors and the log-odds of the outcome. Specifically, for a one-unit increase in a predictor x_i , the log-odds of the outcome (i.e., $p(y = 1)$) versus non-occurrence increase by β_i units, while holding other predictors constant. As a consequence, Logistic Regression is easy to interpret through log-odds.

Logistic Regression

Let's write $p(X) = \Pr(Y = 1|X)$ for short and consider using **balance** to predict **default**. Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

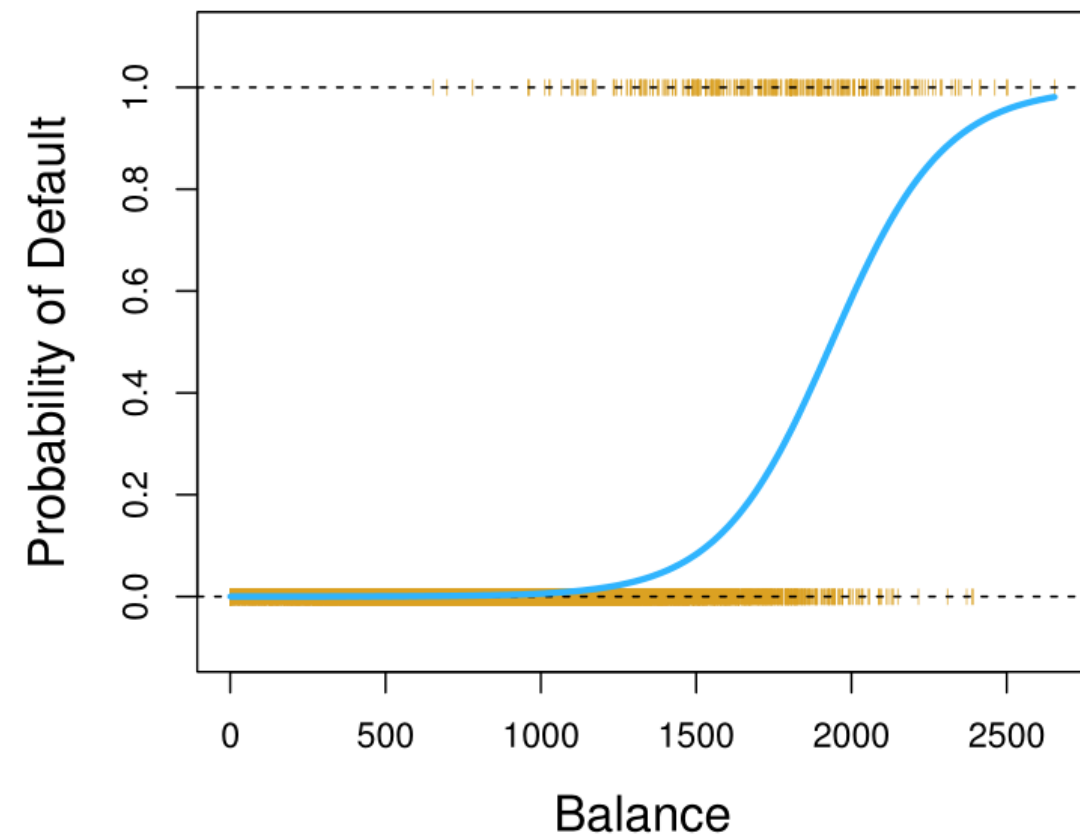
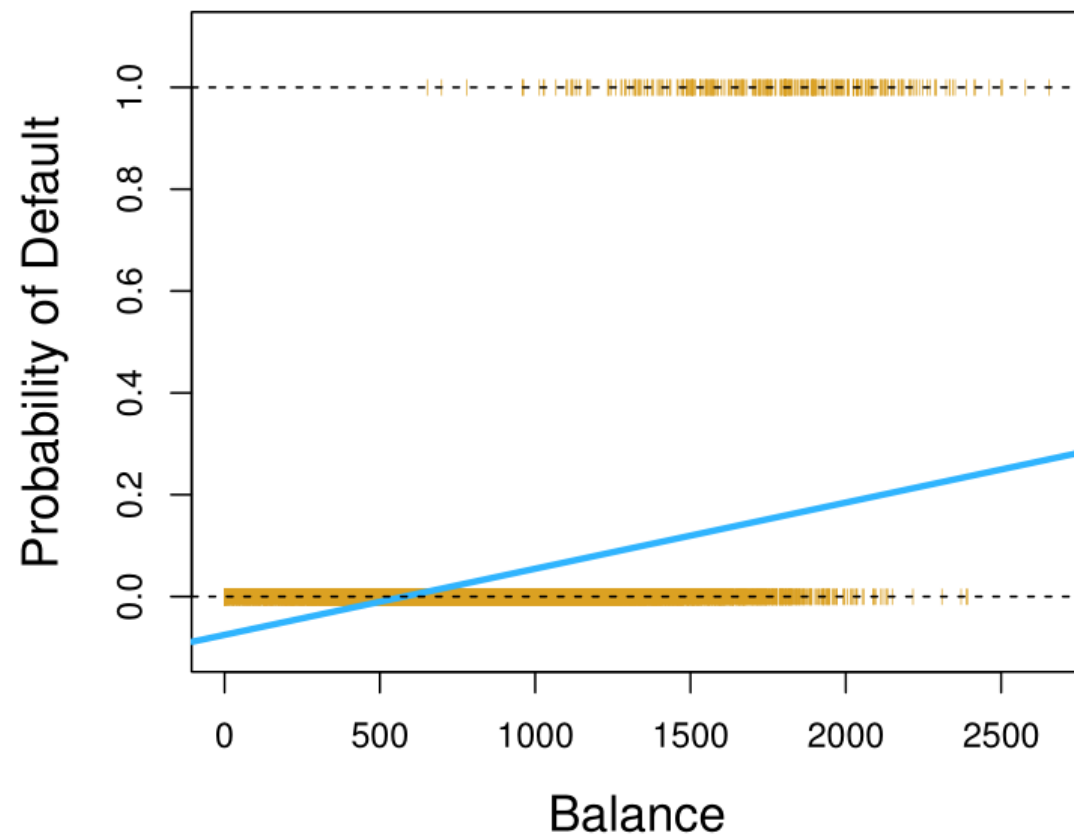
($e \approx 2.71828$ is a mathematical constant [Euler's number.])
It is easy to see that no matter what values β_0 , β_1 or X take, $p(X)$ will have values between 0 and 1.

A bit of rearrangement gives

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X.$$

This monotone transformation is called the *log odds* or *logit* transformation of $p(X)$. (by log we mean *natural log*: $\ln$.)

Linear versus Logistic Regression



Logistic regression ensures that our estimate for $p(X)$ lies between 0 and 1.

Maximum Likelihood

We use maximum likelihood to estimate the parameters.

$$\ell(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

This *likelihood* gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.

Most statistical packages can fit linear logistic regression models by maximum likelihood. In **R** we use the **glm** function.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

Write down the Likelihood more explicitly

Inference for the logistic regression. I

- ▶ The probability function for y is combined with the probability model to obtain:

$$\begin{aligned} f(y_i) &= p^{y_i} (1 - p)^{1-y_i} \\ &= (\Lambda(\beta_1 + \beta_2 x_i))^{y_i} (1 - \Lambda(\beta_1 + \beta_2 x_i))^{1-y_i}. \end{aligned}$$

- ▶ The likelihood function of β_1 and β_2 is

$$L(\beta_1, \beta_2) = \prod_{i=1}^N f(y_i) = \prod_{i=1}^N (\Lambda(\beta_1 + \beta_2 x_i))^{y_i} (1 - \Lambda(\beta_1 + \beta_2 x_i))^{1-y_i}.$$

- ▶ The *maximum likelihood estimates* maximize the *log-likelihood function*:

$$\hat{\beta}_1, \hat{\beta}_2 = \arg \max_{\beta_1, \beta_2} \ln L(\beta_1, \beta_2).$$

From the aspect of Loss function

Machine Learning and FinTech

└ Binary classification

└ Neural networks

List of Loss functions

Loss function	Form
Mean Squared Error (MSE)	$(y - \hat{y})^2$
Binary Crossentropy (BCE)	$-(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))$

In-Class Exercise

- (1) Draw a scatter plot of these three points
- (2) What down the likelihood of the data in the right table $L(\beta_0, \beta_1)$. Assume

$$y|x \sim \text{Bernoulli} \left(\frac{1}{1 + \exp(-(\beta_0 + \beta_1 x))} \right)$$

No	X	Y
1	-1	0
2	2	1
3	2	0

Making Predictions

What is our estimated probability of **default** for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1000}}{1 + e^{-10.6513 + 0.0055 \times 1000}} = 0.006$$

With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 2000}}{1 + e^{-10.6513 + 0.0055 \times 2000}} = 0.586$$

Lets do it again, using **student** as the predictor.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	< 0.0001
student [Yes]	0.4049	0.1150	3.52	0.0004

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{Yes}) = \frac{e^{-3.5041+0.4049 \times 1}}{1 + e^{-3.5041+0.4049 \times 1}} = 0.0431,$$

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{No}) = \frac{e^{-3.5041+0.4049 \times 0}}{1 + e^{-3.5041+0.4049 \times 0}} = 0.0292.$$

`math.log(0.04/0.96) - math.log(0.02/0.98) = 0.4049?`
(Why?)

Logistic regression with several variables

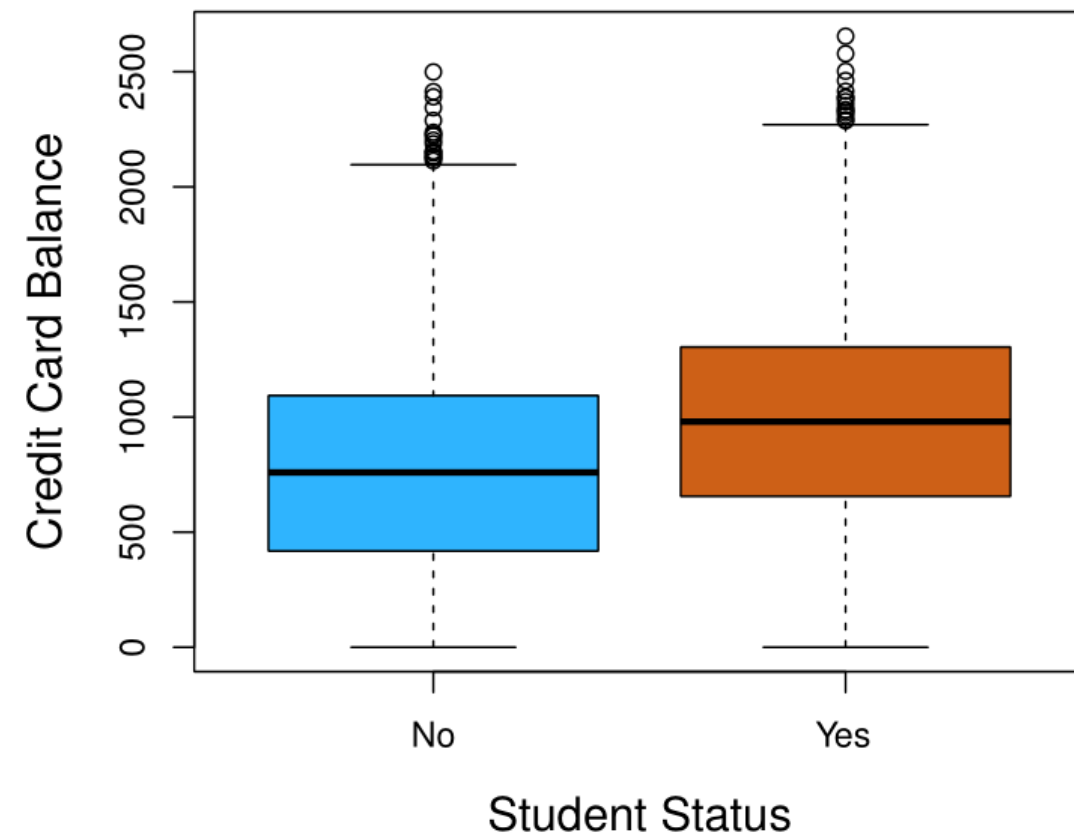
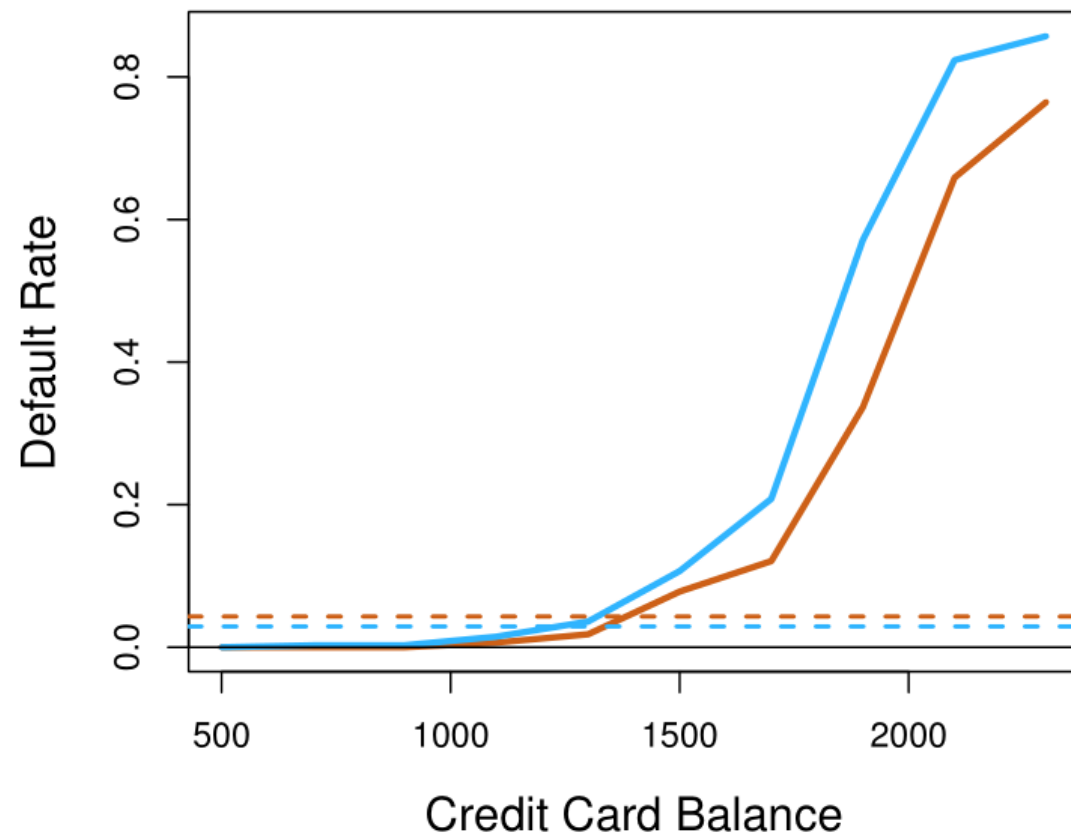
$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student [Yes]	-0.6468	0.2362	-2.74	0.0062

Why is coefficient for **student** negative, while it was positive before?

Confounding



- Students tend to have higher balances than non-students, so their marginal default rate is higher than for non-students.
- But for each level of balance, students default less than non-students.
- Multiple logistic regression can tease this out.

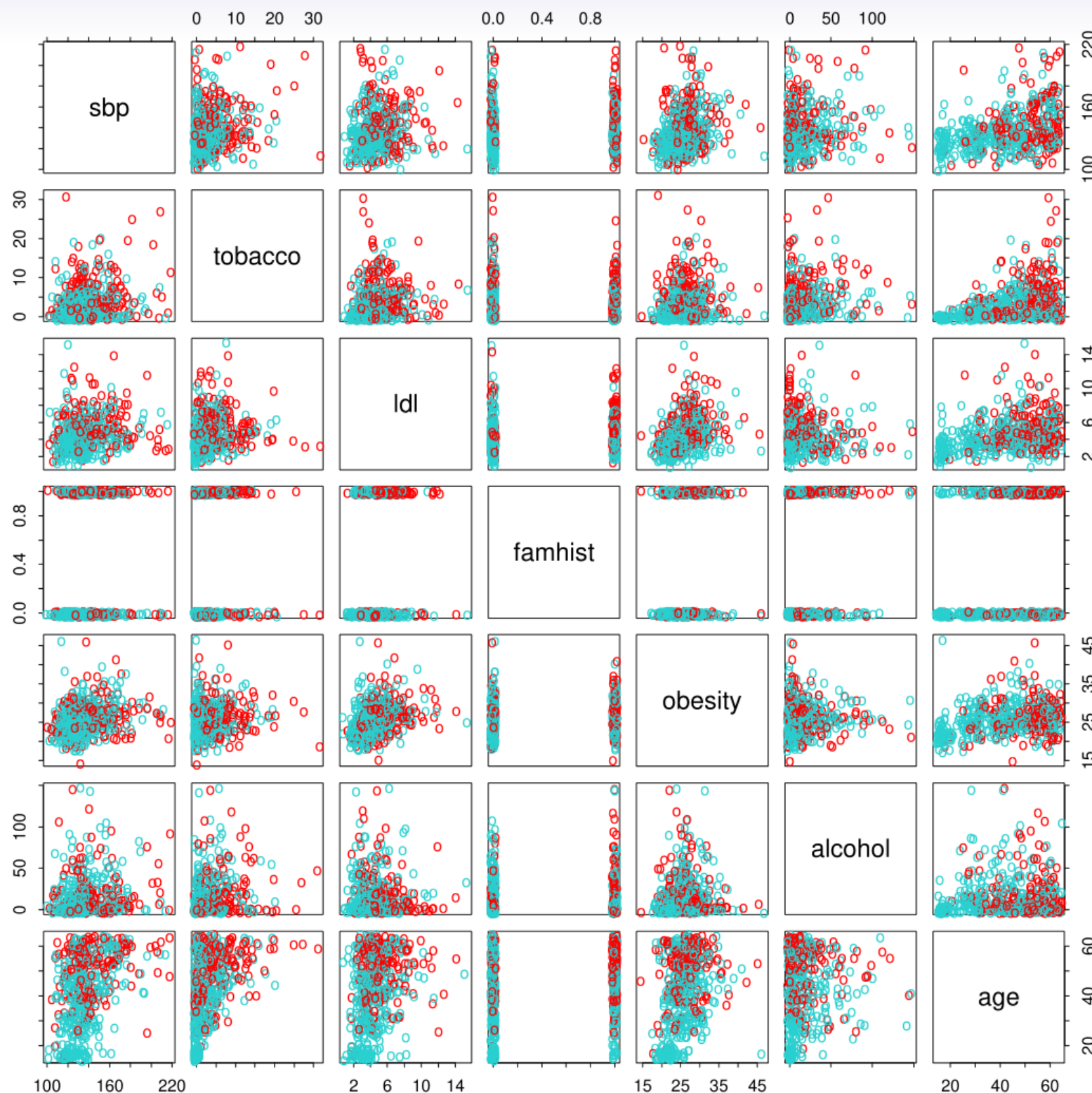
Confounding

- "Confounding" refers to a situation in statistics where a third variable influences both the independent variable and the dependent variable, leading to a spurious association.
- Example: The relationship between a new drug and the recovery rate from a disease.
 - If a confounding factor, such as age, is not controlled for, you might incorrectly conclude that the drug is effective because younger patients (who are more likely to recover anyway due to better overall health) are more likely to use it, rather than older patients.

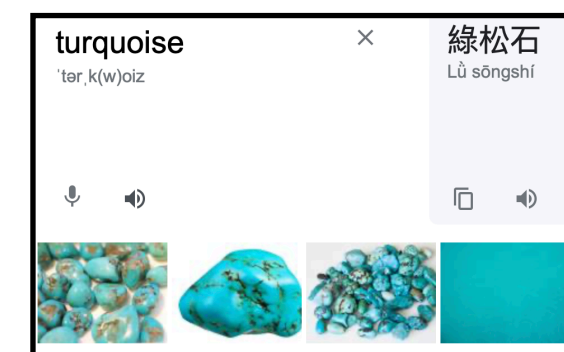
Example: South African Heart Disease

心肌梗塞

- 160 cases of MI (myocardial infarction) and 302 controls (all male in age range 15-64), from Western Cape, South Africa in early 80s.
- Overall prevalence very high in this region: 5.1%.
- Measurements on seven predictors (risk factors), shown in scatterplot matrix.
- Goal is to identify relative strengths and directions of risk factors.
- This was part of an intervention study aimed at educating the public on healthier diets.



Scatterplot matrix of the *South African Heart Disease* data. The response is color coded — The cases (MI) are red, the controls turquoise. **famhist** is a binary variable, with 1 indicating family history of MI.



Logistic Regression with multiple class

- Suppose we have K classes.
- Each class uses parameter vector $\beta_k = (\beta_{0k}, \beta_{1k}, \beta_{2k}, \dots, \beta_{pk})'$
- The sigmoid function $\sigma(t) = \frac{1}{1 + \exp(-t)} = \frac{\exp(t)}{1 + \exp(t)}$
- $\exp(\cdot)$ transfers a number to positive number
- $P(Y = k) \propto \exp(\beta_k' X) = \exp \left(\beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 + \dots + \beta_{pk}X_p \right)$

Logistic regression with more than two classes

So far we have discussed logistic regression with two classes. It is easily generalized to more than two classes. One version (used in the R package **glmnet**) has the symmetric form

$$\Pr(Y = k|X) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p}}{\sum_{\substack{\ell=1 \\ \beta_k = (\beta_{0k}, \beta_{1k}, \dots, \beta_{pk})^\top}}^K e^{\beta_{0\ell} + \beta_{1\ell}X_1 + \dots + \beta_{p\ell}X_p}}$$

Here there is a linear function for *each* class.

(The *mathier* students will recognize that some cancellation is possible, and only $K - 1$ linear functions are needed as in 2-class logistic regression.)

Multiclass logistic regression is also referred to as *multinomial regression*.

Confusion matrix

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Confusion Matrix for Binary Classification

Key components: It breaks down predictions into four categories at a specific threshold: [🔗](#)

- **True Positives (TP):** Correctly predicted positive class. [🔗](#)
- **True Negatives (TN):** Correctly predicted negative class. [🔗](#)
- **False Positives (FP):** Incorrectly predicted as positive (Type I error). [🔗](#)
- **False Negatives (FN):** Incorrectly predicted as negative (Type II error). [🔗](#)

TPR, FNR, FPR, TNR

- True positive rate (TPR) = sensitivity, recall, hit rate

$$\text{TPR} = \frac{\text{TP}}{P}$$

- False negative rate (FNR)= missing rate

$$\text{TNR} = \frac{\text{TN}}{N}$$

- False positive rate (FPR), fall-out

$$\text{FPR} = \frac{\text{FP}}{N}$$

- True negative rate (TNR)

$$\text{TNR} = \frac{\text{TN}}{N}$$

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

P N

Useful measures

accuracy (ACC)

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{P} + \text{N}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- Precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

P N

F1 score

is the **harmonic mean** of **precision** and **sensitivity**:

$$F_1 = 2 \times \frac{\text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

$\hat{P}(Y = 1 | X)$, given a threshold $\Rightarrow \hat{Y}$

Depending on threshold

- TPR, FNR, FPR, TNR
- Accuracy
- Precision
- F1-score

Threshold free:

ROC curve

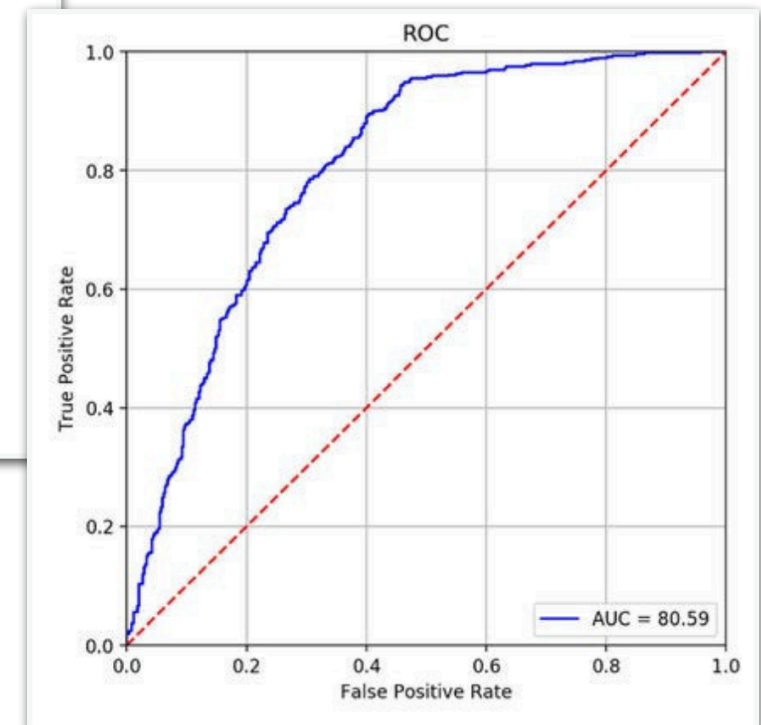
AUC

ROC curve

Google AI review

- **What it is:** A graph that plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. [🔗](#)
- **Axes:**
 - **Y-axis:** True Positive Rate (Sensitivity or Recall), calculated as $TP/(TP + FN)$.
 - **X-axis:** False Positive Rate, calculated as $FP/(FP + TN)$. [🔗](#)
- **Key metric:** The Area Under the Curve (AUC). [🔗](#)
 - An AUC of 1.0 is a perfect classifier.
 - An AUC of 0.5 is a random classifier.
 - The higher the AUC, the better the model's ability to distinguish between classes. [🔗](#)
- **Best for:** Evaluating a classifier's performance across all possible thresholds to understand its overall ability to discriminate between classes. [🔗](#)

$$TPR = \frac{TP}{TP + FN} = \frac{TP}{P}$$
$$FPR = \frac{FP}{FN + TN} = \frac{FP}{N}$$



Thresold = 0.5

Predicted Probability	True Label	Predicted Label
0.9	1	1
0.85	1	1
0.8	0	1
0.7	1	1
0.6	0	1
0.55	0	1
0.5	1	1
0.4	0	0
0.3	0	0
0.1	0	0

	Predicted Positive (1)	Predicted Negative (0)
Actual Positive (1)	TP: 4	FN: 0
Actual Negative (0)	FP: 3	TN: 3

Threshold = 0.8

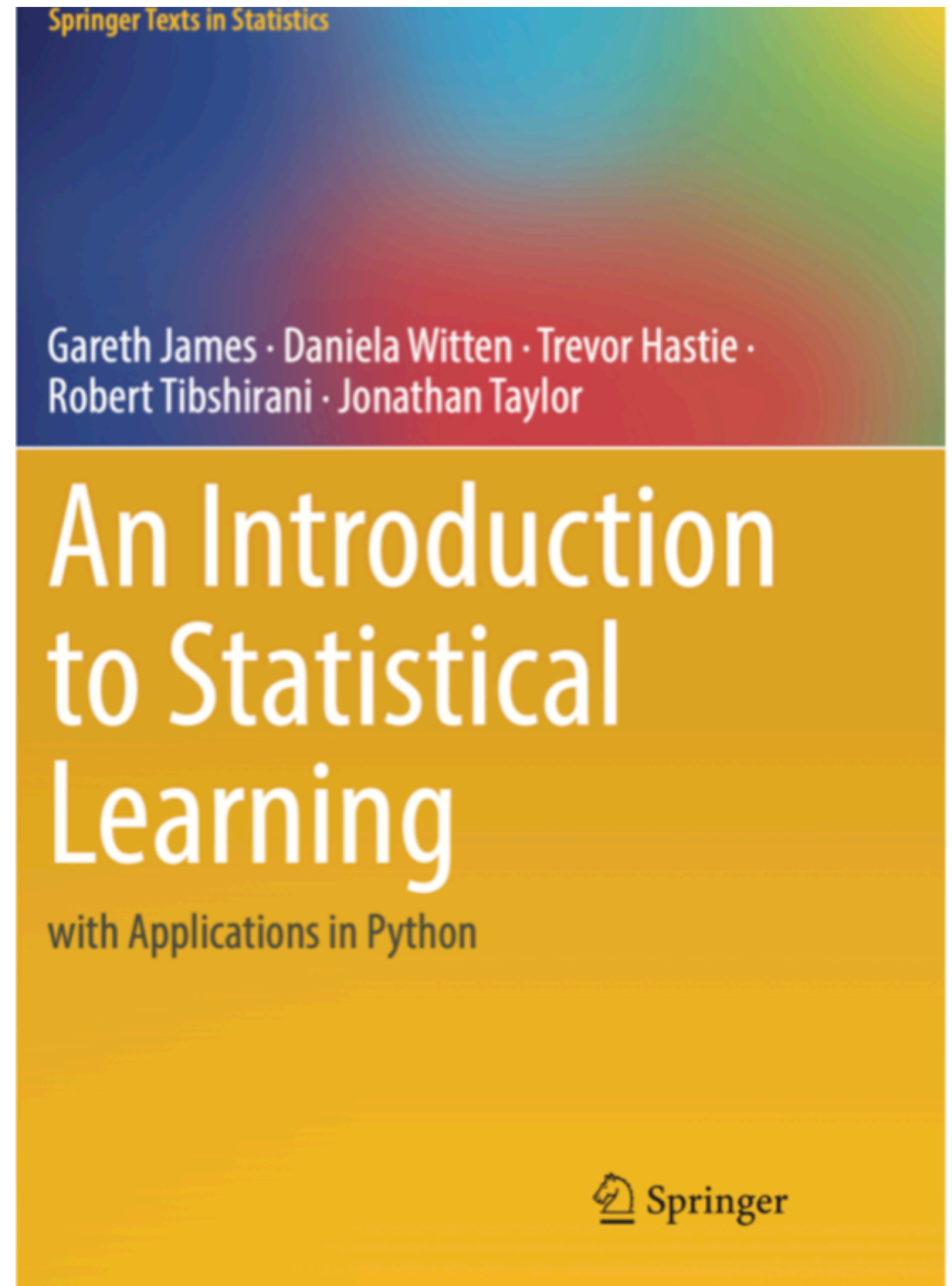
Predicted Probability	True Label	Predicted Label
0.9	1	1
0.85	1	1
0.8	0	1
0.7	1	0
0.6	0	0
0.55	0	0
0.5	1	0
0.4	0	0
0.3	0	0
0.1	0	0

	Predicted Positive (1)	Predicted Negative (0)
Actual Positive (1)	TP: 2	FN: 2
Actual Negative (0)	FP: 1	TN: 5

In-Class exercise

- $\hat{\beta}'X = [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]$
- Predicted Probabilities:
 - $[0.9, 0.85, 0.8, 0.7, 0.6, 0.55, 0.5, 0.4, 0.3, 0.1]$
- True Labels: $[1, 1, 0, 1, 0, 0, 1, 0, 0, 0]$
- Draw a scatter plot of $\hat{p}$ vs $\hat{\beta}'X$
- Set the threshold = 0.5, find the confusion matrix, TPR, FPR.
- Set the threshold = 0.8, find the confusion matrix, TPR, FPR.
- For simplicity, consider the threshold = 0, 0.2, 0.4, ..., 1.
 - Approximate the ROC curve and calculate the AUC.

4.4 Generative Models for Classification



<https://www.statlearning.com/>

Discriminant Analysis

Here the approach is to model the distribution of X in each of the classes separately, and then use *Bayes theorem* to flip things around and obtain $\Pr(Y|X)$.

When we use normal (Gaussian) distributions for each class, this leads to linear or quadratic discriminant analysis.

However, this approach is quite general, and other distributions can be used as well. We will focus on normal distributions.

1701-1761 (aged 59), London, England

The Reverend
Thomas Bayes



Portrait purportedly of Bayes used in a 1936 book,^[1] but it is doubtful whether the portrait is actually of him.^[2] No earlier portrait or claimed portrait survives.

Alma mater University of Edinburgh

Known for Bayesian statistics
Bayes' theorem
Conditional probability
Inverse probability
Bayes prior
Bayes factor
Bayesian inference
See full list

Scientific career

Fields Probability

Signature

T. Bayes.

Bayes theorem for classification

Thomas Bayes was a famous mathematician whose name represents a big subfield of statistical and probabilistic modeling. Here we focus on a simple result, known as Bayes theorem:

$$\Pr(Y = k|X = x) = \frac{\Pr(X = x|Y = k) \cdot \Pr(Y = k)}{\Pr(X = x)}$$

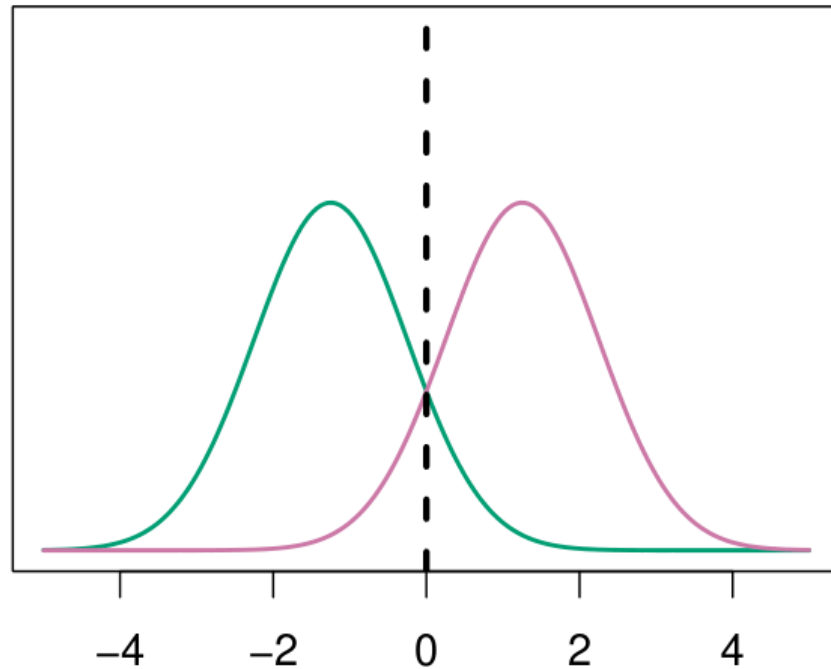
One writes this slightly differently for discriminant analysis:

$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}, \quad \text{where}$$

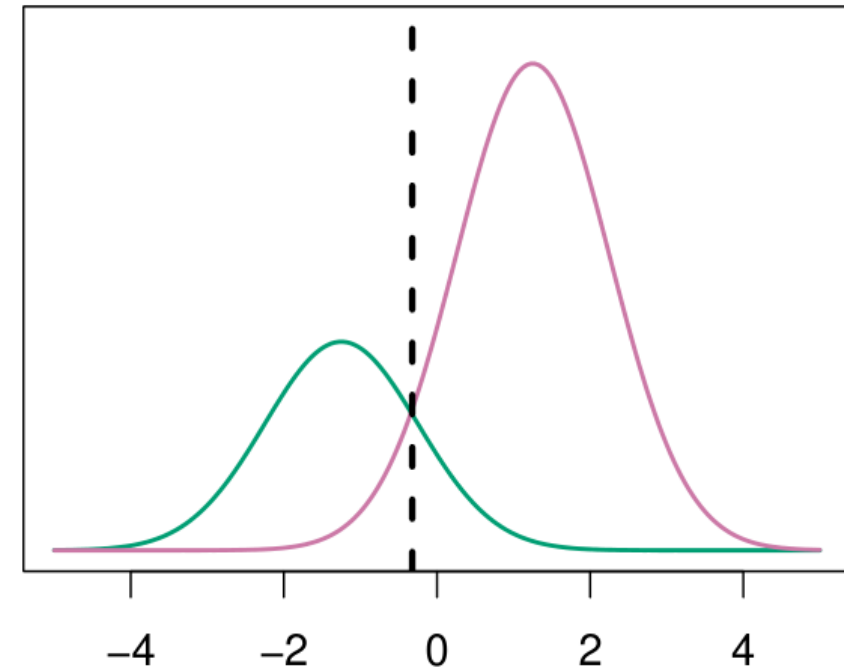
- $f_k(x) = \Pr(X = x|Y = k)$ is the *density* for X in class k . Here we will use normal densities for these, separately in each class.
- $\pi_k = \Pr(Y = k)$ is the marginal or *prior* probability for class k .

Classify to the highest density

$$\pi_1=.5, \quad \pi_2=.5$$



$$\pi_1=.3, \quad \pi_2=.7$$



We classify a new point according to which density is highest.

When the priors are different, we take them into account as well, and compare $\pi_k f_k(x)$. On the right, we favor the pink class — the decision boundary has shifted to the left.

Law of total probability

Proof

1. The events $\{Y = 1\}, \{Y = 2\}, \dots, \{Y = K\}$ form a **partition** of the sample space: exactly one of them happens.
2. For any event about X , in particular the event $\{X = x\}$, we can decompose it along that partition:

$$\Pr(X = x) = \sum_{k=1}^K \Pr(X = x, Y = k).$$

This is just "split the probability according to which class occurred."

3. Apply the product rule to each term:

$$\Pr(X = x, Y = k) = \Pr(X = x \mid Y = k) \Pr(Y = k).$$

4. Substitute into the sum:

$$\Pr(X = x) = \sum_{k=1}^K \Pr(X = x \mid Y = k) \Pr(Y = k).$$

5. Using the snapshot notation $\Pr(X = x \mid Y = k) = f_k(x)$ and $\Pr(Y = k) = \pi_k$, we get

$$\Pr(X = x) = \sum_{k=1}^K \pi_k f_k(x).$$

Why discriminant analysis?

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

Linear Discriminant Analysis when $p = 1$

The Gaussian density has the form

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

Here μ_k is the mean, and σ_k^2 the variance (in class k). We will assume that all the $\sigma_k = \sigma$ are the same.

Plugging this into Bayes formula, we get a rather complex expression for $p_k(x) = \Pr(Y = k|X = x)$:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma}\right)^2}}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_l}{\sigma}\right)^2}}$$

Happily, there are simplifications and cancellations.

Discriminant functions

To classify at the value $X = x$, we need to see which of the $p_k(x)$ is largest. Taking logs, and discarding terms that do not depend on k , we see that this is equivalent to assigning x to the class with the largest *discriminant score*:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Note that $\delta_k(x)$ is a *linear* function of x .

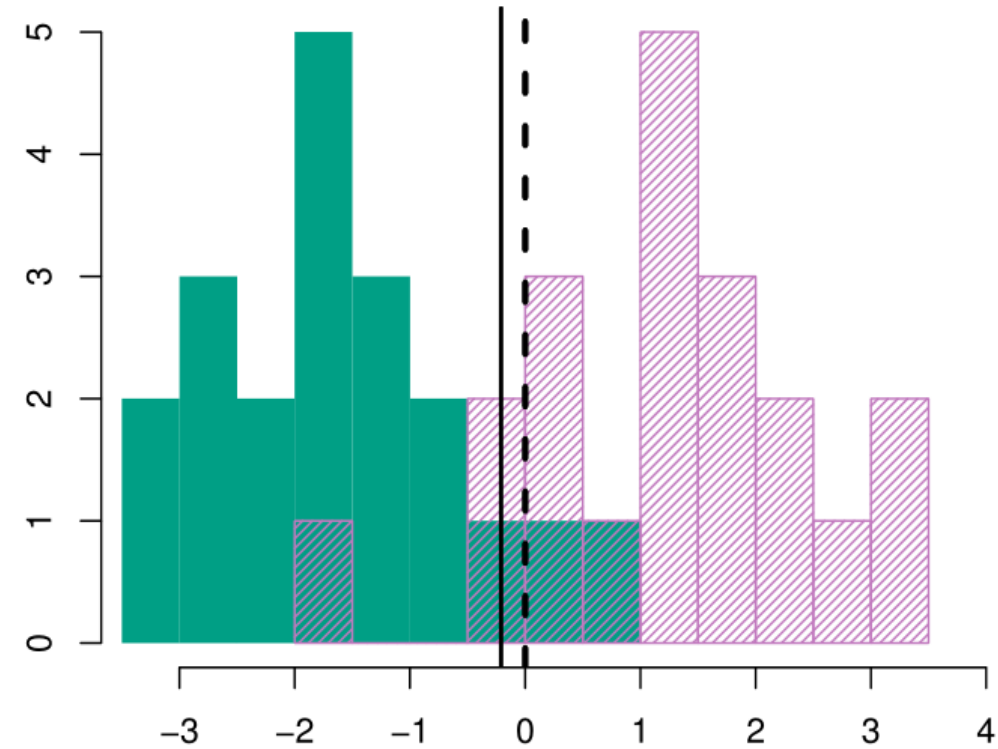
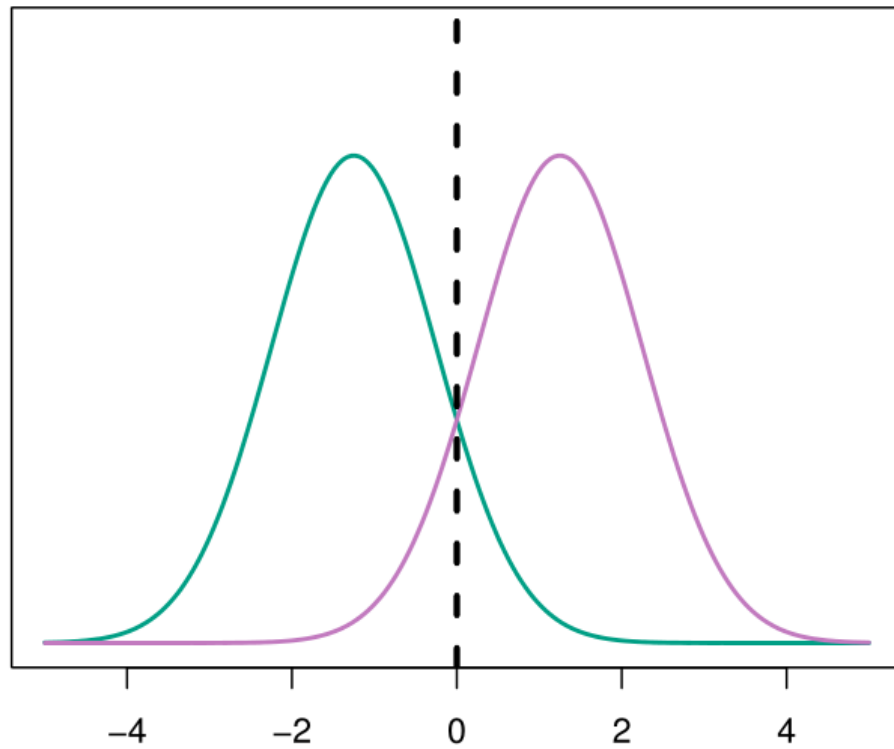
If there are $K = 2$ classes and $\pi_1 = \pi_2 = 0.5$, then one can see that the *decision boundary* is at

$$x = \frac{\mu_1 + \mu_2}{2}.$$

(See if you can show this)

In-Class exercise

- $\begin{cases} \delta_1(x) = ? \\ \delta_2(x) = ? \end{cases}$
- Let $\delta_1(x) = \delta_2(x)$, show that math simplifications yields:
$$x = \frac{\mu_1 + \mu_2}{2}$$



Example with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, and $\sigma^2 = 1$.

Typically we don't know these parameters; we just have the training data. In that case we simply estimate the parameters and plug them into the rule.

Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{n}$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i=k} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n - K} \sum_{k=1}^K \sum_{i: y_i=k} (x_i - \hat{\mu}_k)^2$$

$$= \sum_{k=1}^K \frac{n_k - 1}{n - K} \cdot \hat{\sigma}_k^2$$

where $\hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i: y_i=k} (x_i - \hat{\mu}_k)^2$ is the usual formula for the estimated variance in the k th class.

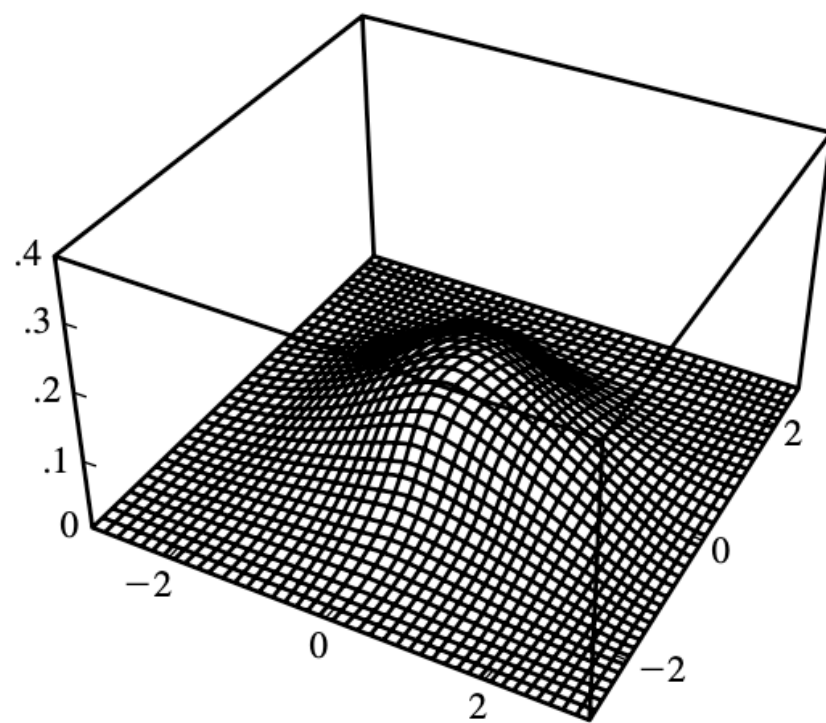
LDA when $p > 1$: Multivariate Normal Density

The bivariate normal density is

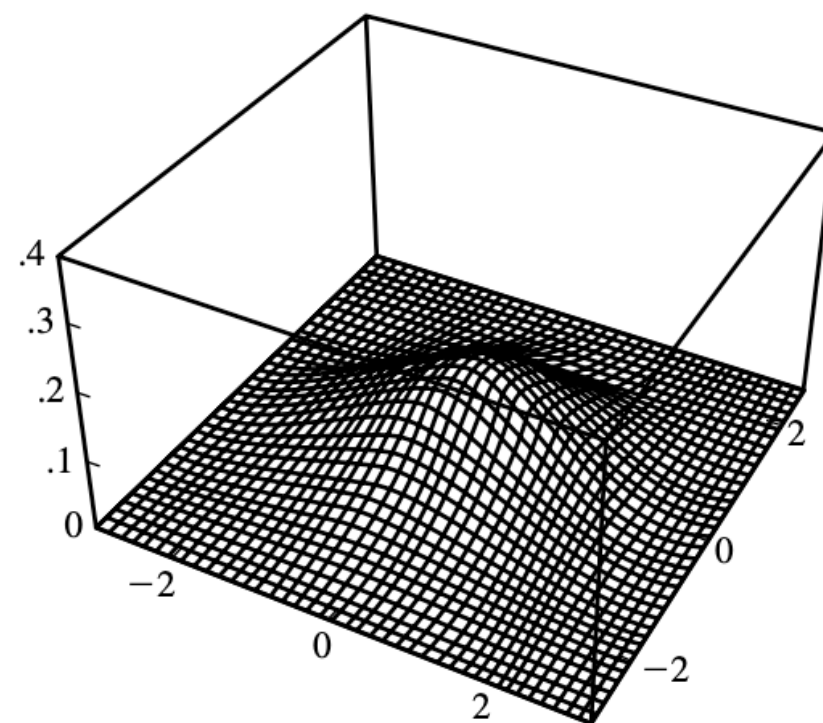
$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]\right),$$

for $-\infty \leq x, y \leq \infty$.

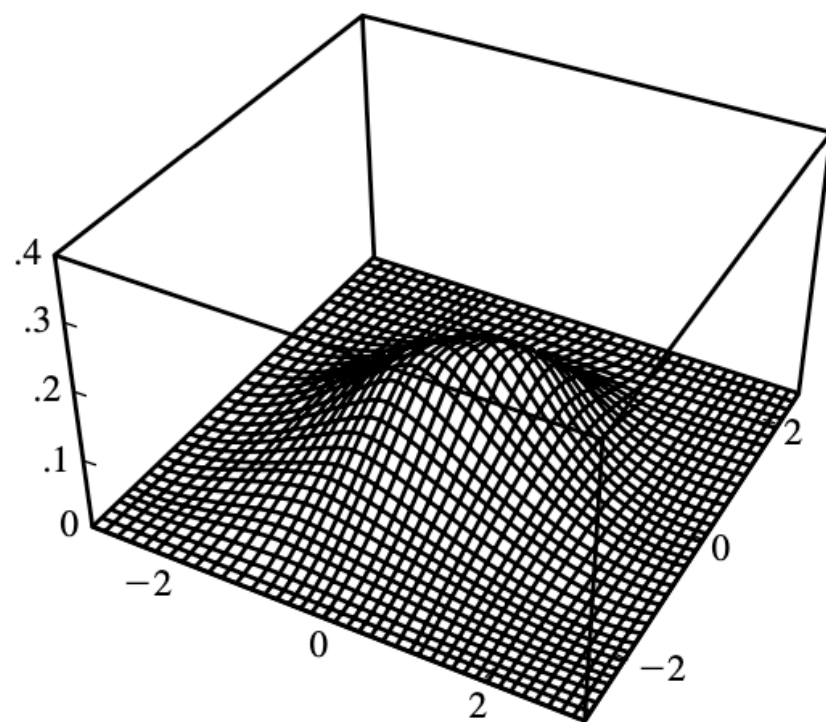
Denote $\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}\right)$



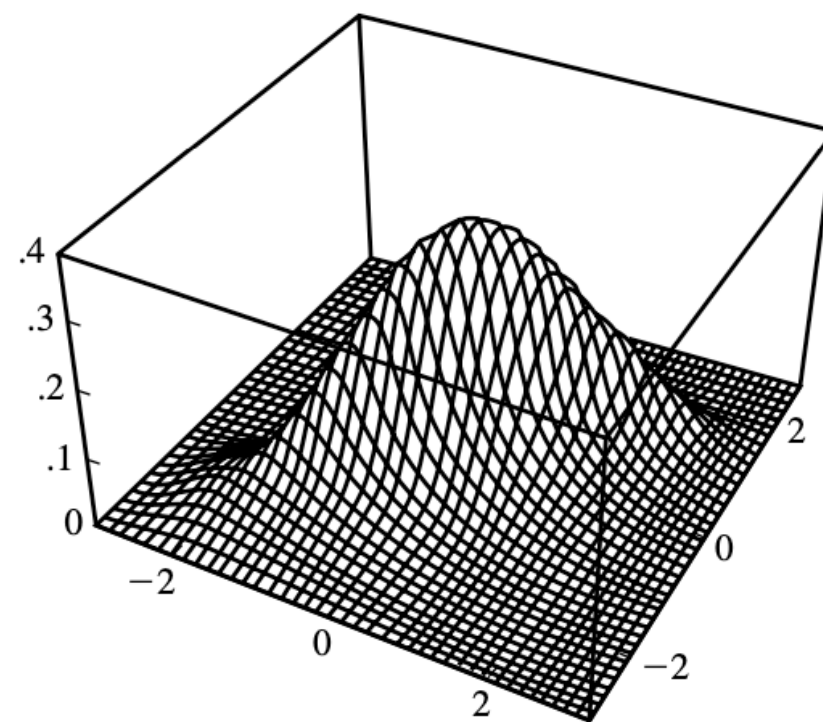
(a)



(b)



(c)



(d)

FIGURE 3.9 Bivariate normal densities with $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ and (a) $\rho = 0$, (b) $\rho = .3$, (c) $\rho = .6$, (d) $\rho = .9$.

Understand the contour lines of the density:

$$\frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X\sigma_Y} = \text{constant}.$$

See a contour for example.

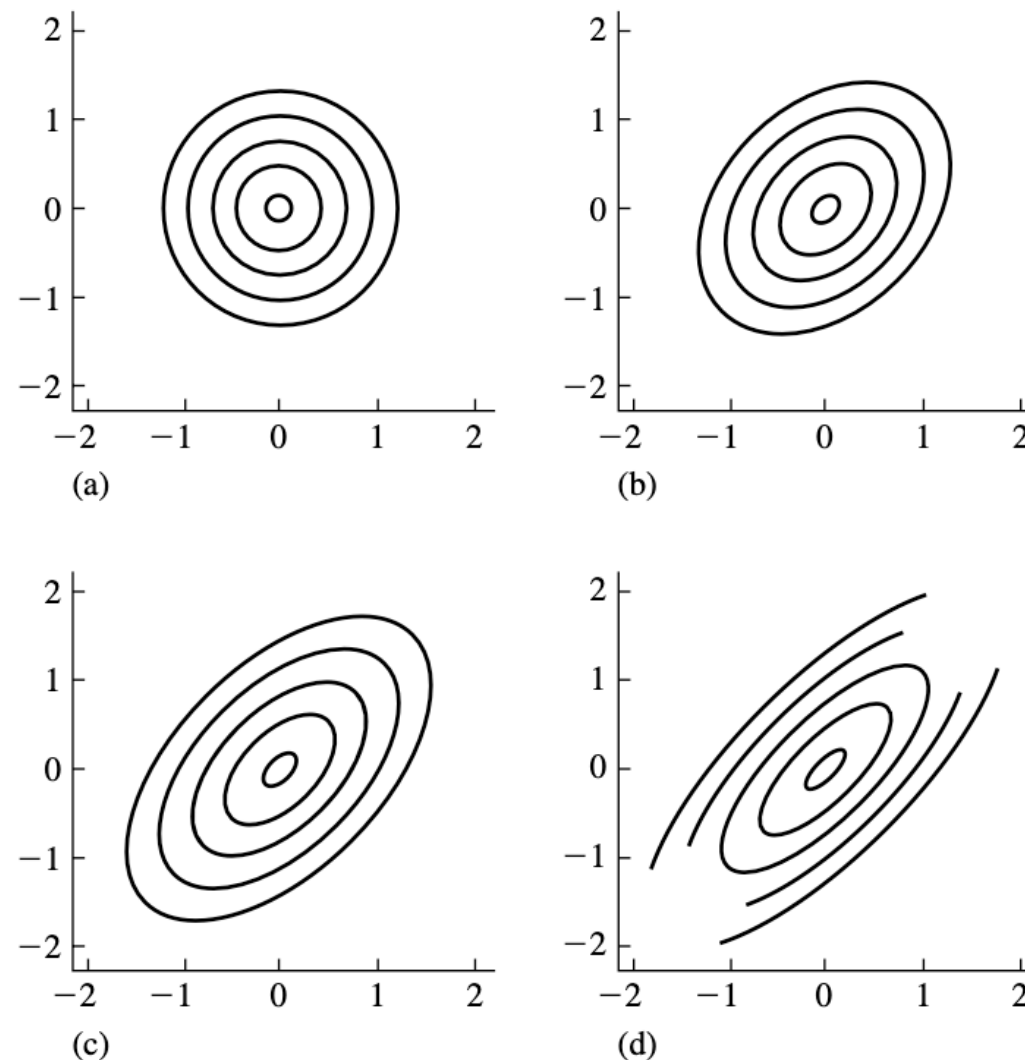


FIGURE 3.10 The elliptical contours of the bivariate normal densities of Figure 3.9.

$$X \sim N_d(\mu, \Sigma)$$

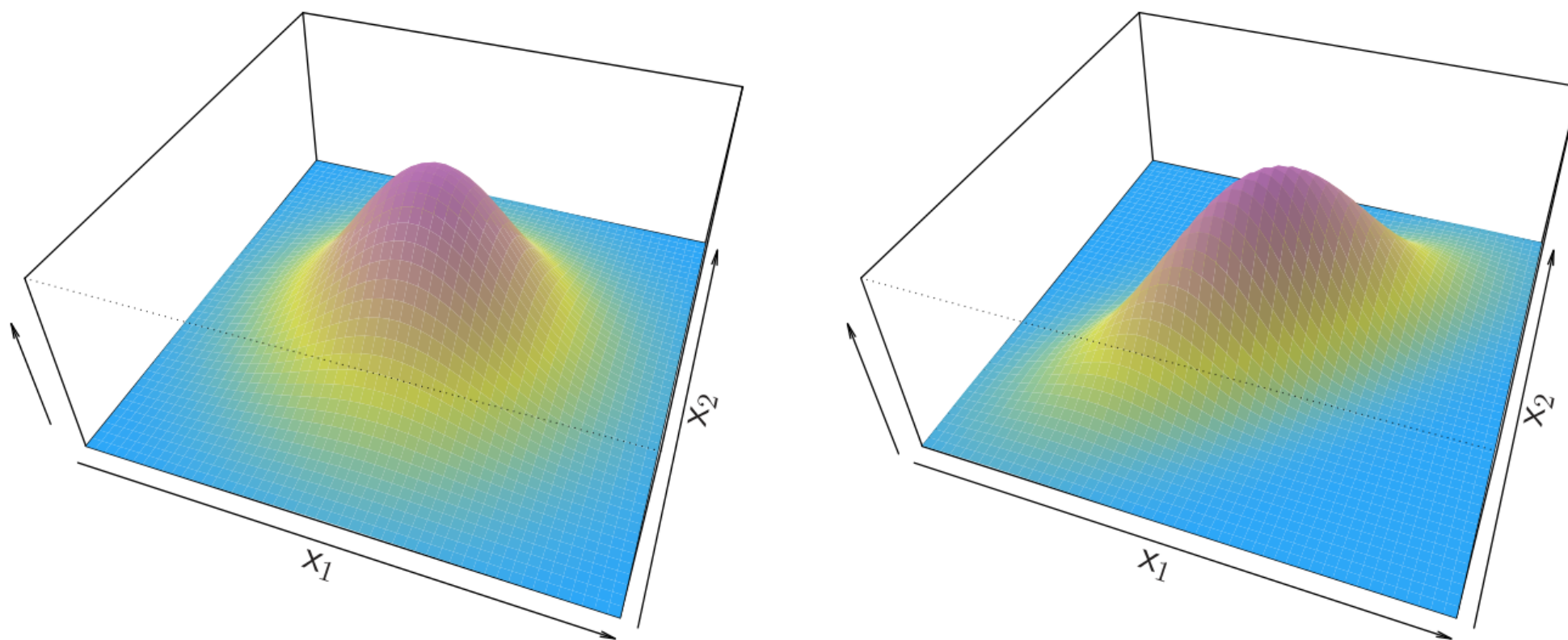
The multivariate normal density of $X \sim N_d(\mu, \Sigma)$ is

$$f(x) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{(x - \mu)' \Sigma^{-1} (x - \mu)}{2}\right),$$

for $-\infty \leq x_i \leq \infty$, $i = 1, \dots, d$.

Show how the contour is affected by Σ using linear algebra!

Linear Discriminant Analysis when $p > 1$



$$\text{Density: } f(x) = \frac{1}{(2\pi)^{p/2} |\mathbf{\Sigma}|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T \mathbf{\Sigma}^{-1}(x-\mu)}$$

$$\text{Discriminant function: } \delta_k(x) = x^T \mathbf{\Sigma}^{-1} \mu_k - \frac{1}{2} \mu_k^T \mathbf{\Sigma}^{-1} \mu_k + \log \pi_k$$

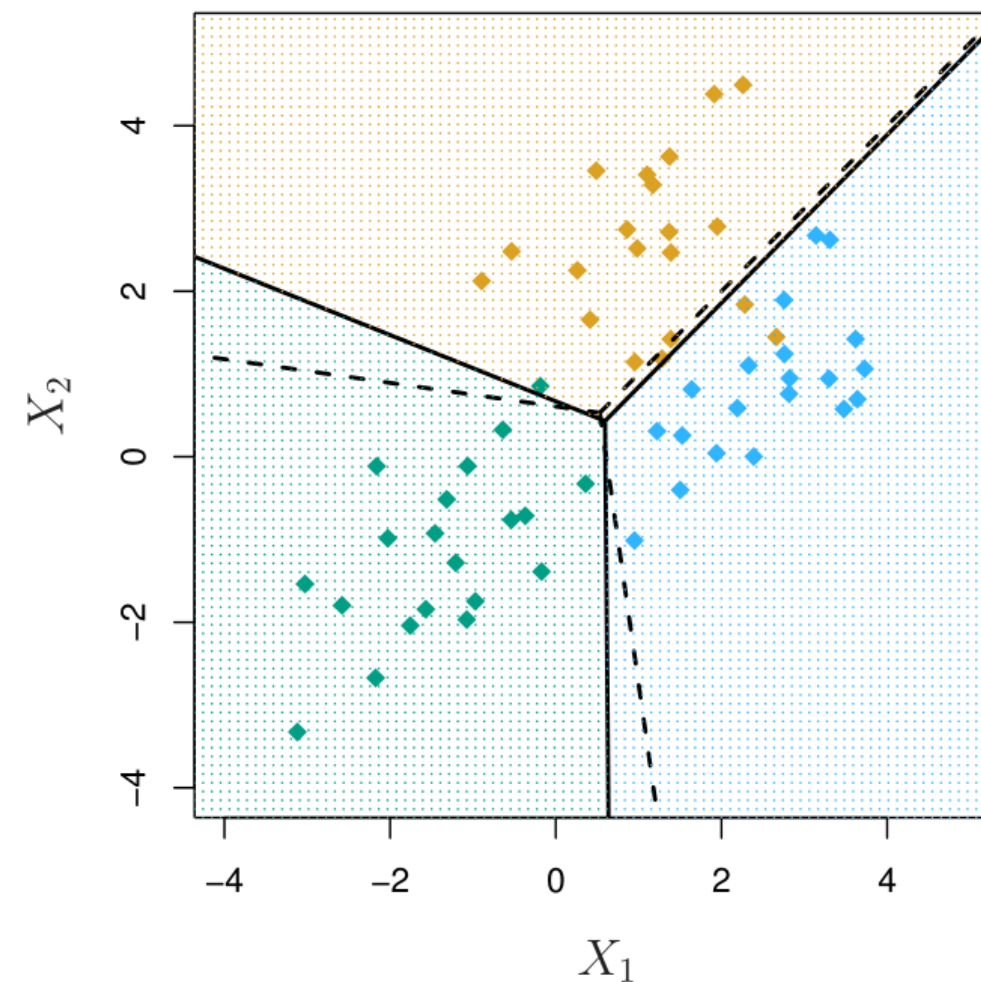
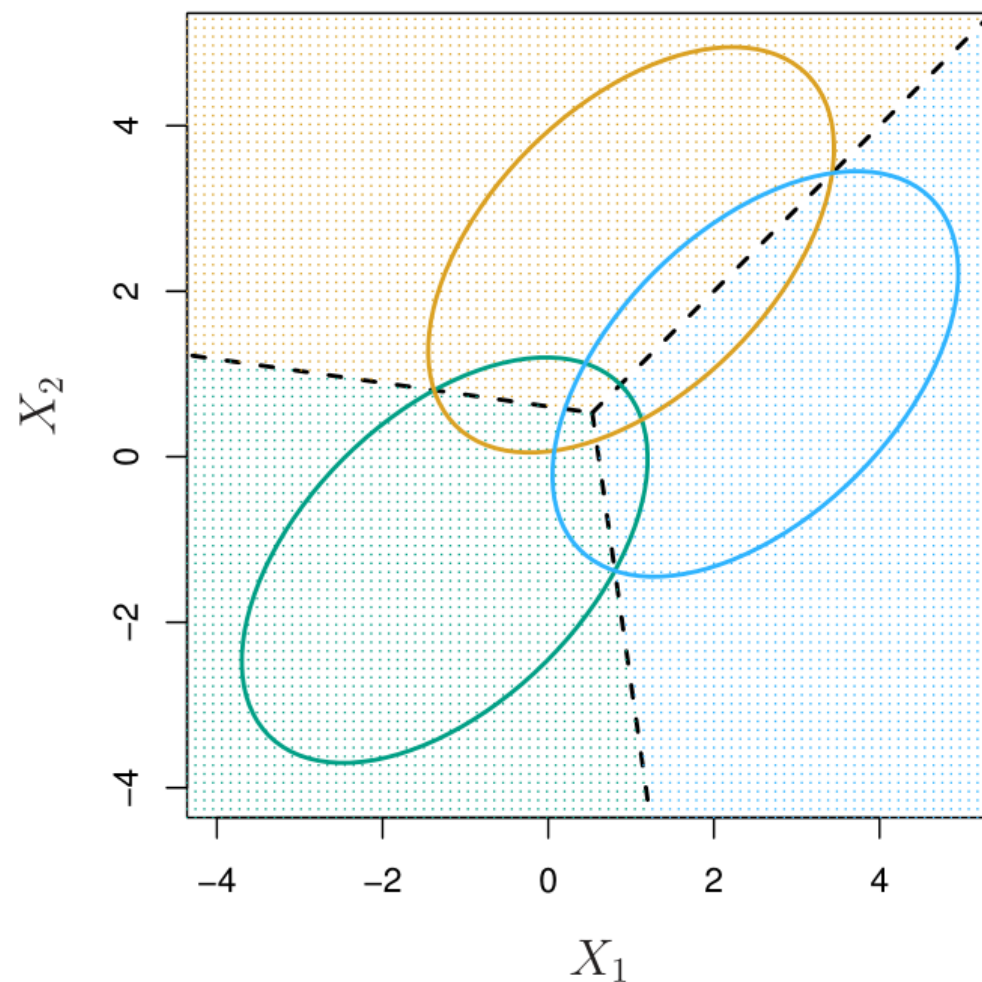
Despite its complex form,

$\delta_k(x) = c_{k0} + c_{k1}x_1 + c_{k2}x_2 + \dots + c_{kp}x_p$ — a linear function.

In-Class Exercise

- Derive the discriminant function
- Show why the discrimination function is a linear function

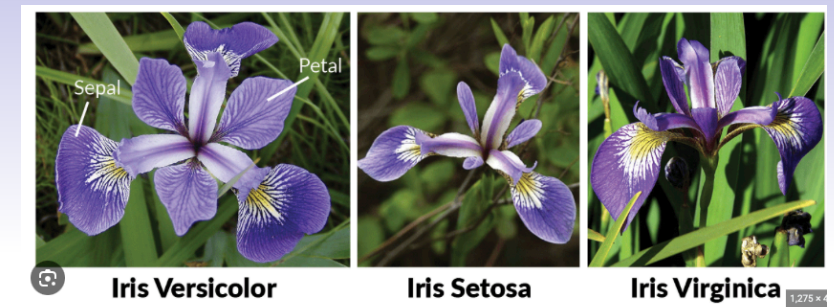
Illustration: $p = 2$ and $K = 3$ classes



Here $\pi_1 = \pi_2 = \pi_3 = 1/3$.

The dashed lines are known as the *Bayes decision boundaries*. Were they known, they would yield the fewest misclassification errors, among all possible classifiers.

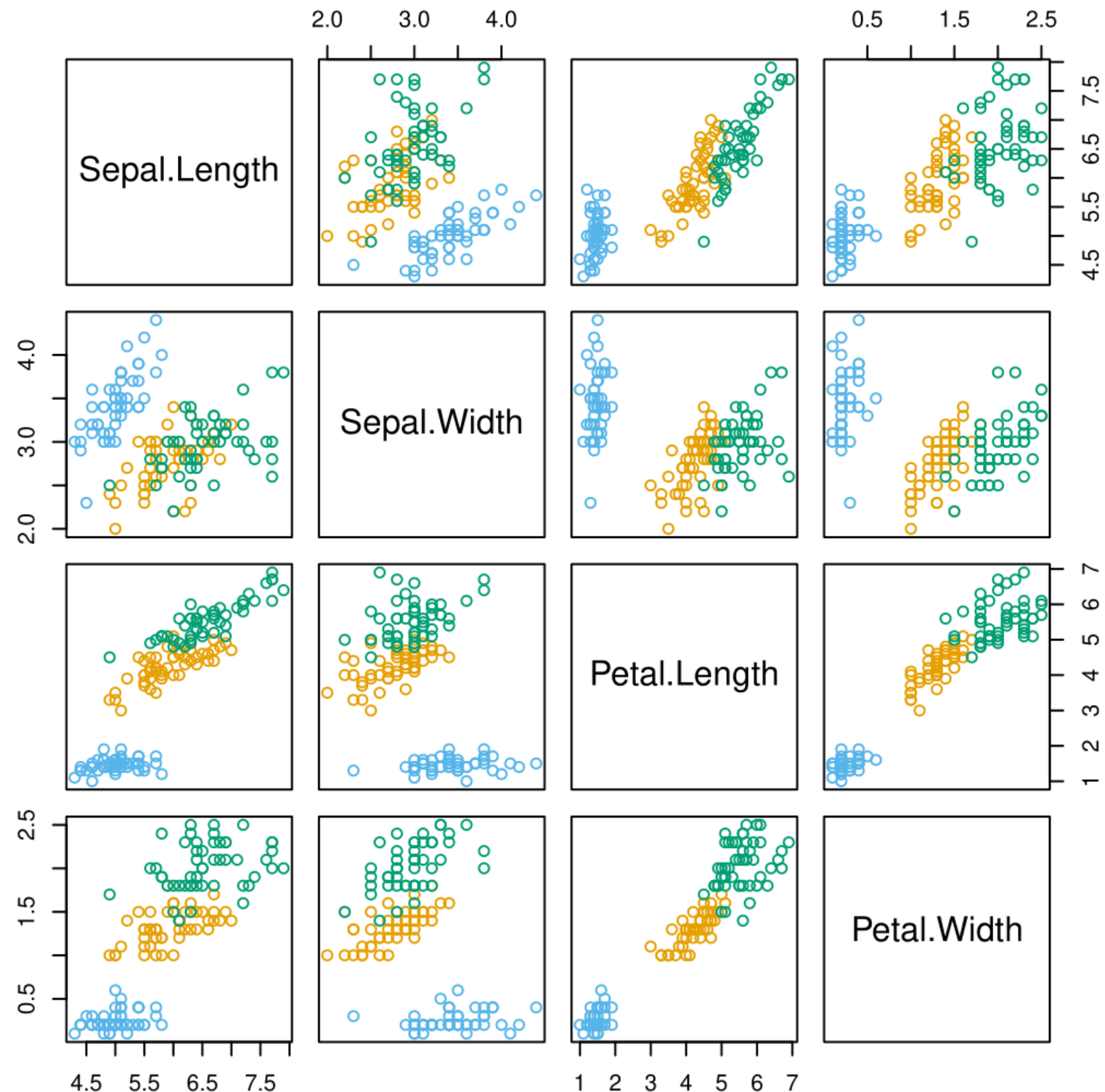
Fisher's Iris Data



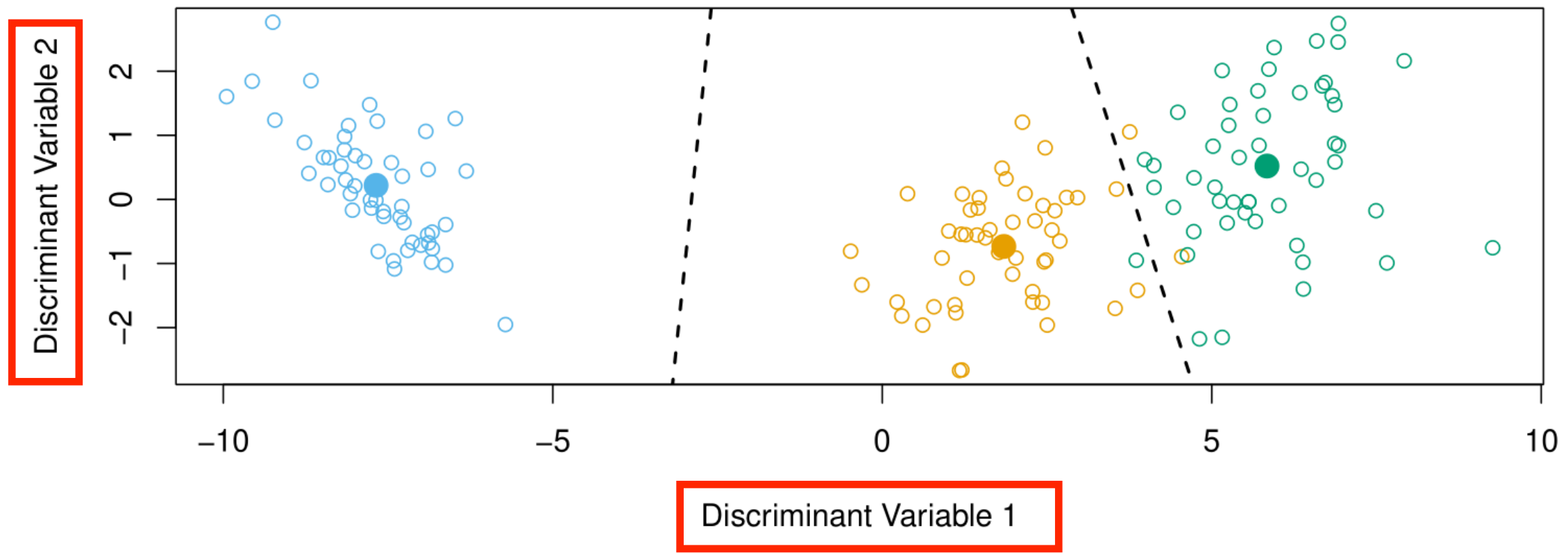
4 variables
3 species
50 samples/class

- Setosa
- Versicolor
- Virginica

LDA classifies all
but 3 of the 150
training samples
correctly.



Fisher's Discriminant Plot



When there are K classes, linear discriminant analysis can be viewed exactly in a $K - 1$ dimensional plot.

Why? Because it essentially classifies to the closest centroid, and they span a $K - 1$ dimensional plane.

Even when $K > 3$, we can find the “best” 2-dimensional plane for visualizing the discriminant rule.

From $\delta_k(x)$ to probabilities

Once we have estimates $\hat{\delta}_k(x)$, we can turn these into estimates for class probabilities:

$$\widehat{\Pr}(Y = k|X = x) = \frac{e^{\hat{\delta}_k(x)}}{\sum_{l=1}^K e^{\hat{\delta}_l(x)}}.$$

So classifying to the largest $\hat{\delta}_k(x)$ amounts to classifying to the class for which $\widehat{\Pr}(Y = k|X = x)$ is largest.

When $K = 2$, we classify to class 2 if $\widehat{\Pr}(Y = 2|X = x) \geq 0.5$, else to class 1.

LDA on Credit Data

What is the FPR?
Waht is the FNR?

		<i>True Default Status</i>		
		No N	Yes P	Total
<i>Predicted Default Status</i>	N No	9644	252	9896
	P Yes	23	81	104
	Total	9667	333	10000

$(23 + 252)/10000$ errors — a 2.75% misclassification rate!

Some caveats:

- This is *training* error, and we may be overfitting. Not a big concern here since $n = 10000$ and $p = 2$!
- If we classified to the prior — always to class **No** in this case — we would make $333/10000$ errors, or only 3.33%.
- Of the true **No**'s, we make $23/9667 = 0.2\%$ errors; of the true **Yes**'s, we make $252/333 = 75.7\%$ errors!

Types of errors

Check in class!

False positive rate: The fraction of negative examples that are classified as positive — 0.2% in example.

False negative rate: The fraction of positive examples that are classified as negative — 75.7% in example.

We produced this table by classifying to class **Yes** if

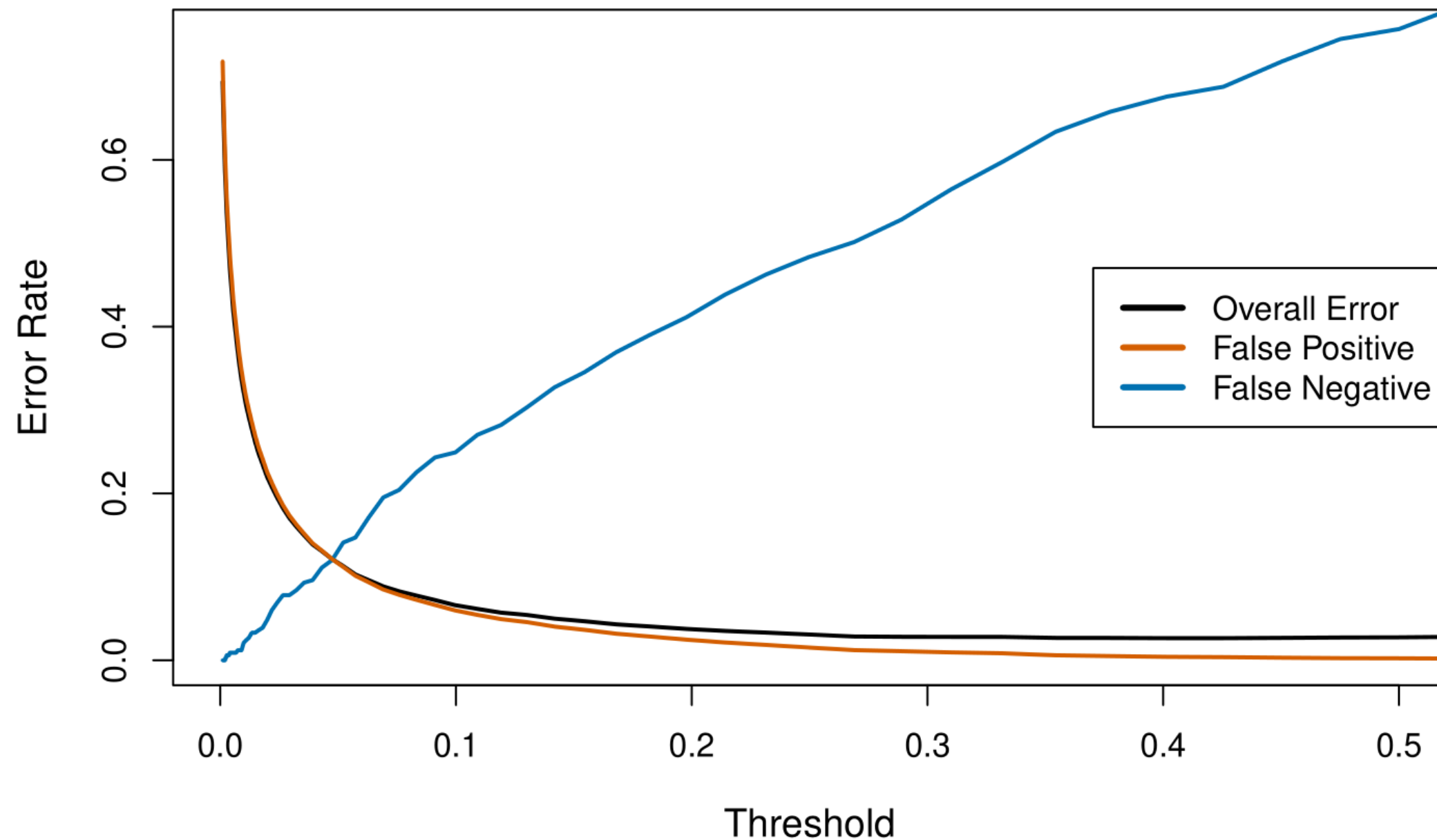
$$\widehat{\text{Pr}}(\text{Default} = \text{Yes} | \text{Balance}, \text{Student}) \geq 0.5$$

We can change the two error rates by changing the threshold from 0.5 to some other value in $[0, 1]$:

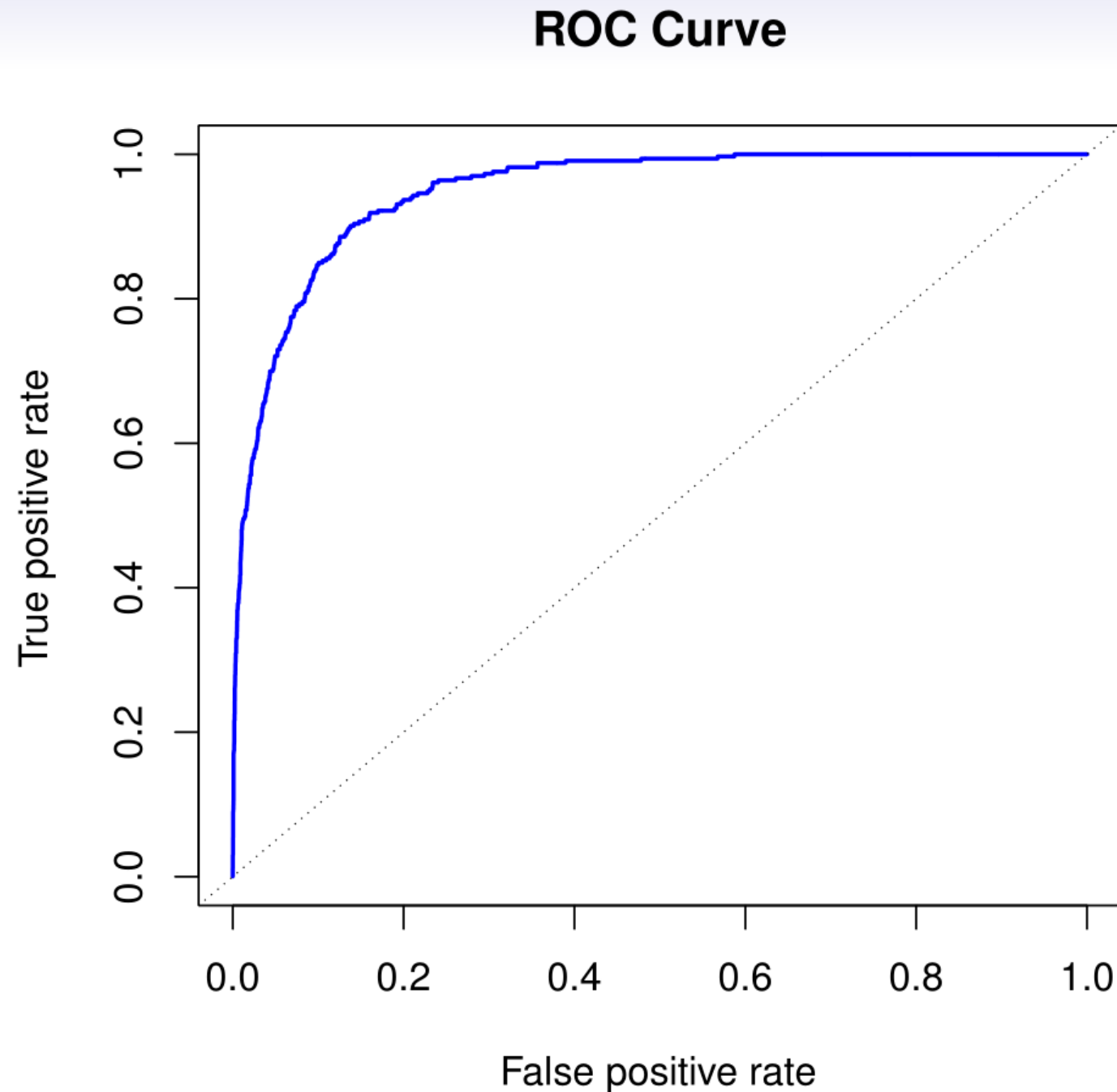
$$\widehat{\text{Pr}}(\text{Default} = \text{Yes} | \text{Balance}, \text{Student}) \geq \textit{threshold},$$

and vary *threshold*.

Varying the *threshold*



In order to reduce the false negative rate, we may want to reduce the threshold to 0.1 or less.



The *ROC plot* displays both simultaneously.

Sometimes we use the *AUC* or *area under the curve* to summarize the overall performance. Higher *AUC* is good.

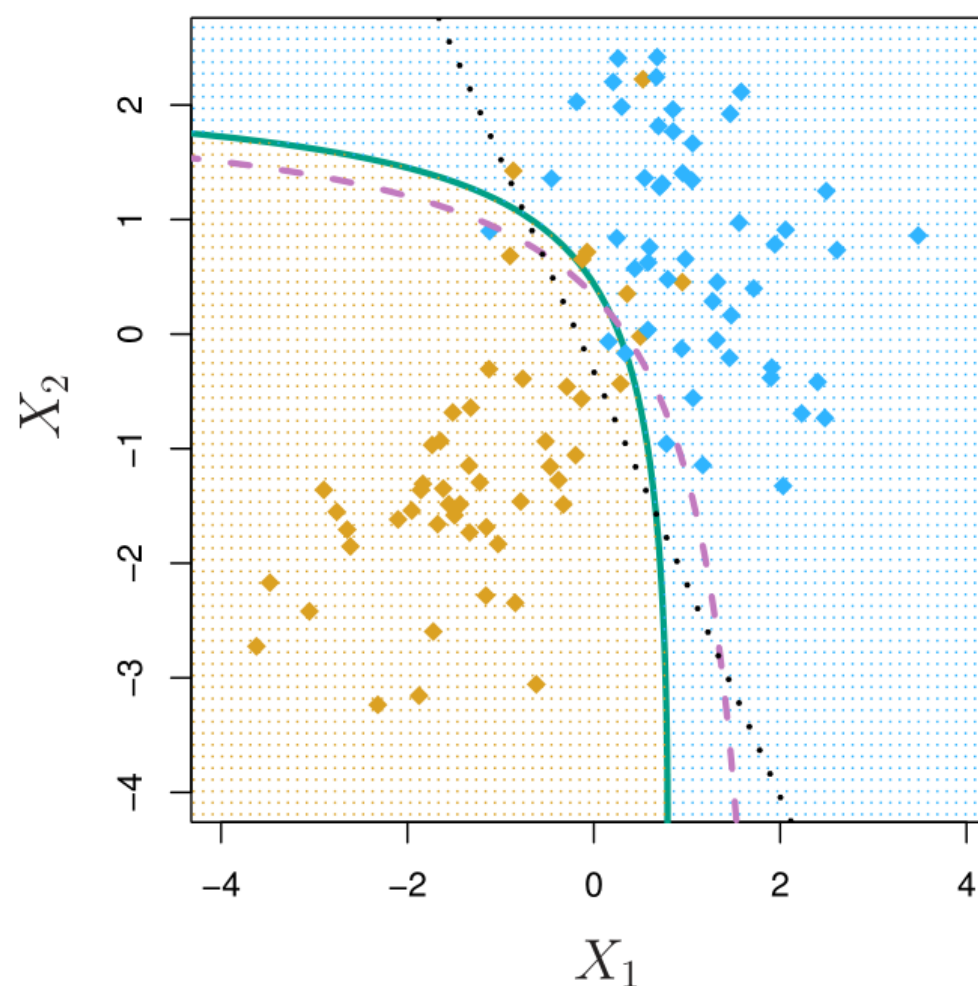
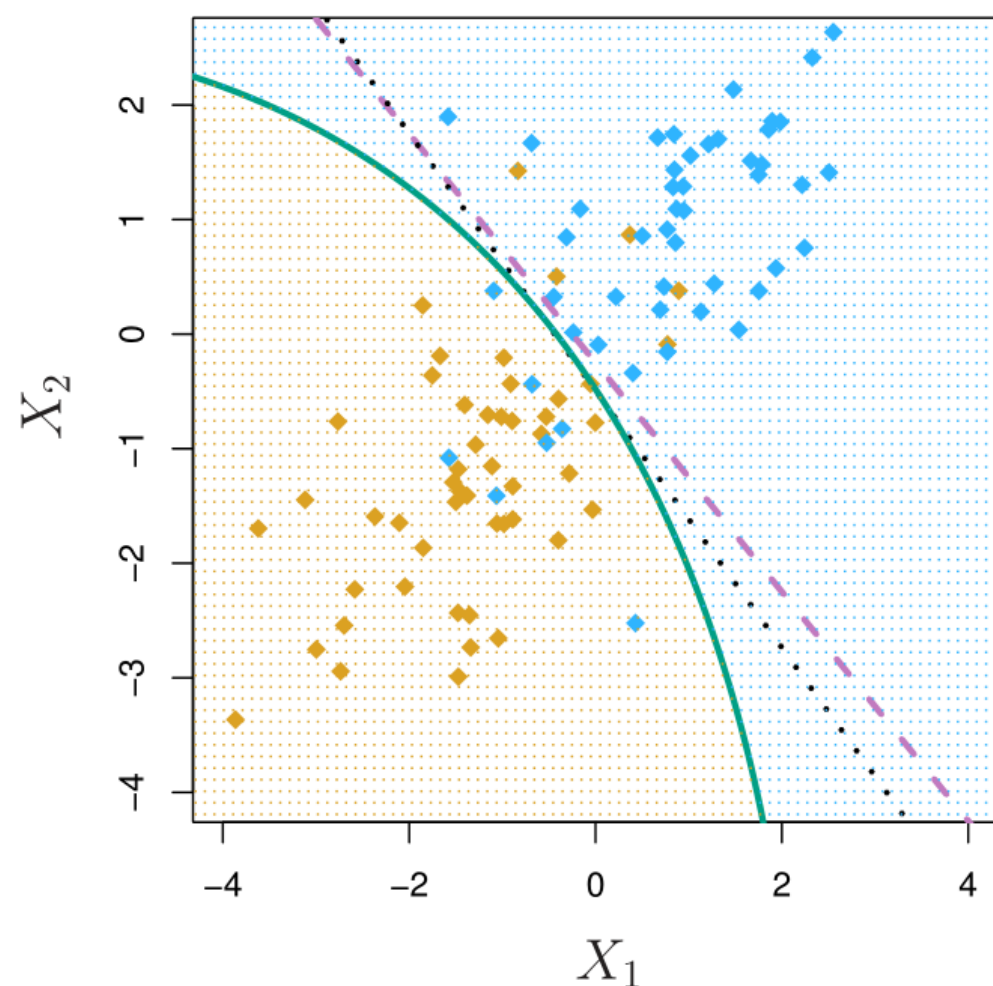
Other forms of Discriminant Analysis

$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

When $f_k(x)$ are Gaussian densities, with the same covariance matrix Σ in each class, this leads to linear discriminant analysis. By altering the forms for $f_k(x)$, we get different classifiers.

- With Gaussians but different Σ_k in each class, we get *quadratic discriminant analysis*.
- With $f_k(x) = \prod_{j=1}^p f_{jk}(x_j)$ (conditional independence model) in each class we get *naive Bayes*. For Gaussian this means the Σ_k are diagonal.
- Many other forms, by proposing specific density models for $f_k(x)$, including nonparametric approaches.

Quadratic Discriminant Analysis



$$\delta_k(x) = -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k) + \log \pi_k - \frac{1}{2} \log |\Sigma_k|$$

Because the Σ_k are different, the quadratic terms matter.

Naive Bayes

Assumes features are independent in each class.

Useful when p is large, and so multivariate methods like QDA and even LDA break down.

- Gaussian naive Bayes assumes each Σ_k is diagonal:

$$\begin{aligned}\delta_k(x) &\propto \log \left[\pi_k \prod_{j=1}^p f_{kj}(x_j) \right] \\ &= -\frac{1}{2} \sum_{j=1}^p \left[\frac{(x_j - \mu_{kj})^2}{\sigma_{kj}^2} + \log \sigma_{kj}^2 \right] + \log \pi_k\end{aligned}$$

- can use for *mixed* feature vectors (qualitative and quantitative). If X_j is qualitative, replace $f_{kj}(x_j)$ with probability mass function (histogram) over discrete categories.

Despite strong assumptions, naive Bayes often produces good classification results.

Logistic Regression versus LDA

For a two-class problem, one can show that for LDA

$$\log \left(\frac{p_1(x)}{1 - p_1(x)} \right) = \log \left(\frac{p_1(x)}{p_2(x)} \right) = c_0 + c_1 x_1 + \dots + c_p x_p$$

So it has the same form as logistic regression.

The difference is in how the parameters are estimated.

- Logistic regression uses the conditional likelihood based on $\Pr(Y|X)$ (known as *discriminative learning*).
- LDA uses the full likelihood based on $\Pr(X, Y)$ (known as *generative learning*).
- Despite these differences, in practice the results are often very similar.

Footnote: logistic regression can also fit quadratic boundaries like QDA, by explicitly including quadratic terms in the model.

Logistic Regression including quadratic terms ($p = 1$)

To illustrate how polynomial-transformed features and interactions can enhance Logistic Regression's predictions, consider these two examples. For the first, with a single feature ($p = 1$), we generate 100 data points for x between -1 and 1 . y is set to 1 for x values less than -0.2 or greater than 0.5 , and 0 otherwise. Refer to Fig. 3 for this data representation.

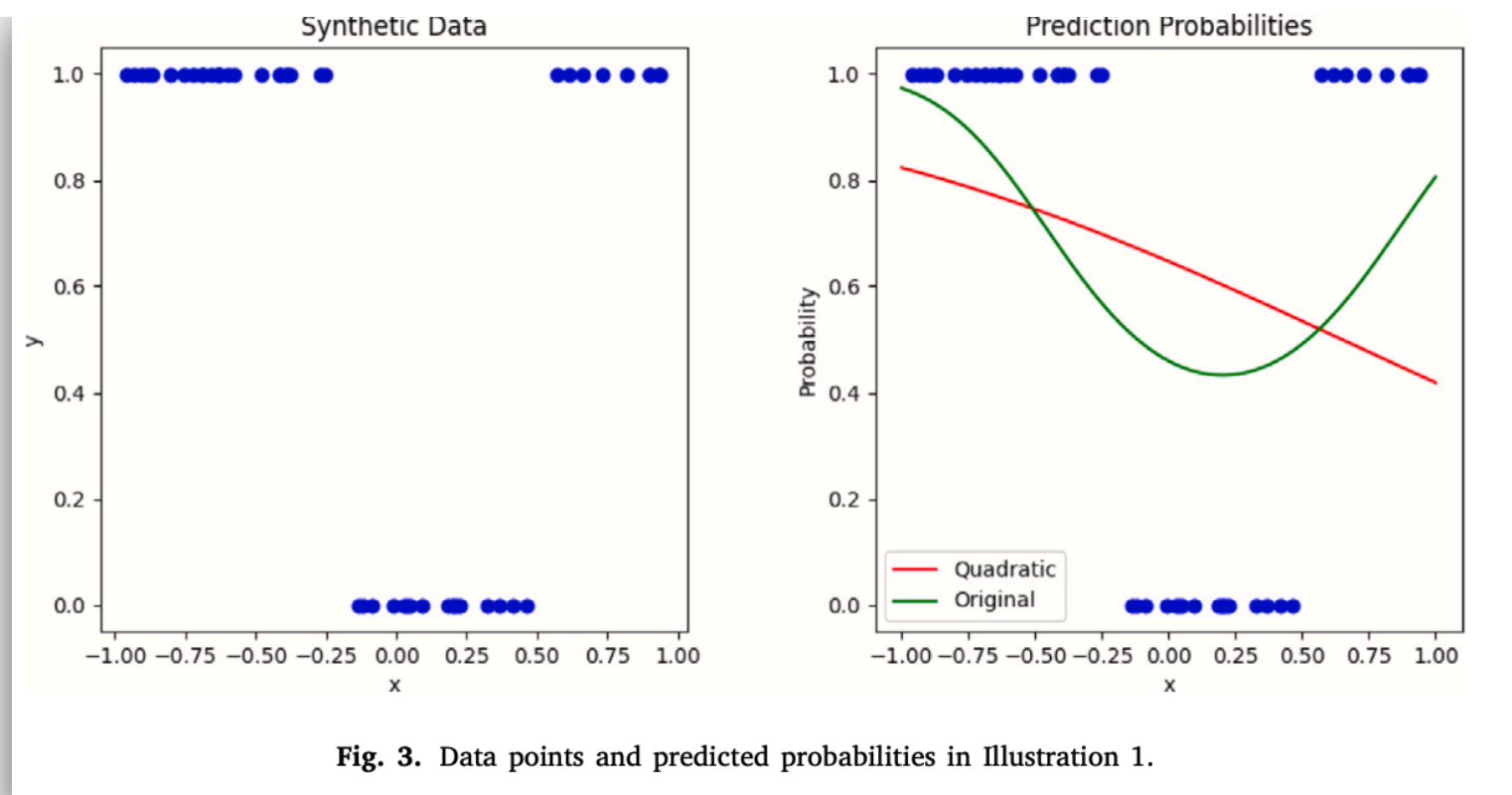


Fig. 3. Data points and predicted probabilities in Illustration 1.

We fit Logistic Regression using the original feature, x , and its quadratic polynomial, x^2 . This quadratic inclusion equips the Logistic

Logistic Regression including quadratic terms ($p = 2$)

In our subsequent illustration with two predictors, x_1 and x_2 , the coordinates of points are stored in X , while y denotes a binary label identifying each point's respective circle. Fig. 4 displays two concentric circles: the inner red diamond points represent $y = 1$, and the outer blue circle points signify $y = 0$.

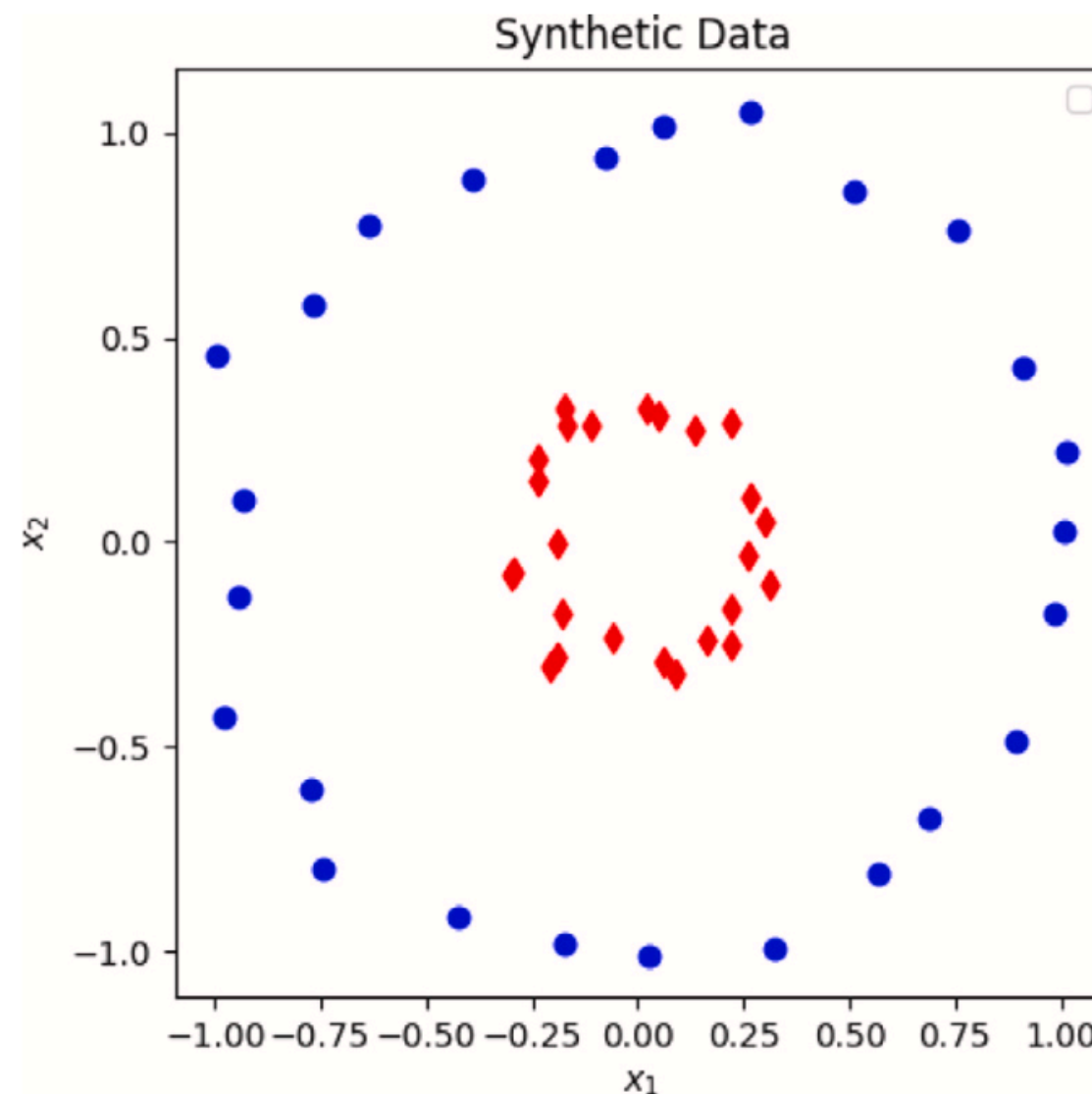


Fig. 4. Data points in Illustration 2.

We trained two Logistic Regression models on this synthetic data: one using only the original features, x_1, x_2 , and another enhanced with quadratic features (x_1, x_2, x_1^2, x_2^2 , and x_1x_2). This inclusion captures non-linearities. The decision boundaries of both models are visualized in Fig. 5. The left plot, based on original features, has a linear boundary

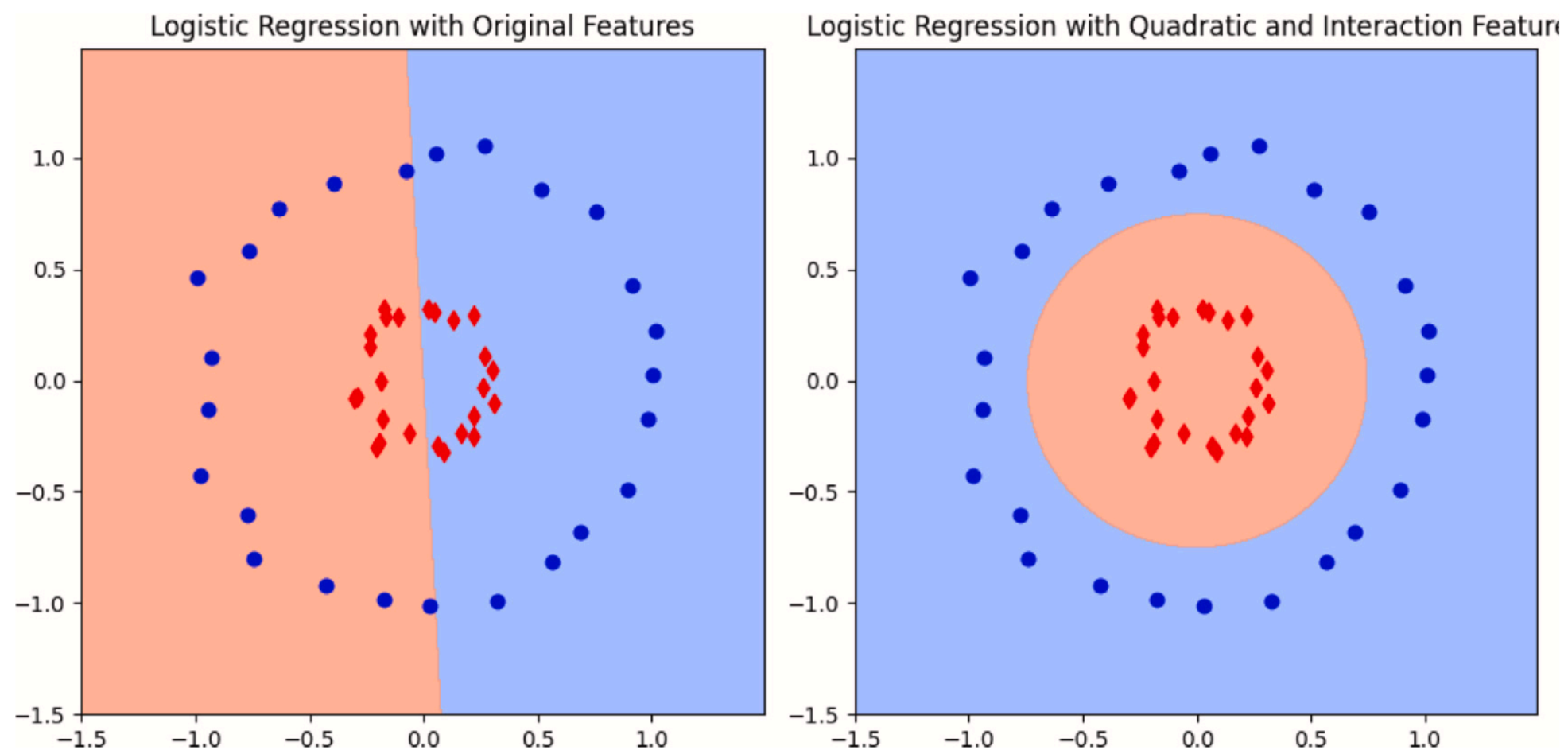


Fig. 5. Decision boundaries in Illustration 2.

Summary

- Logistic regression is very popular for classification, especially when $K = 2$.
- LDA is useful when n is small, or the classes are well separated, and Gaussian assumptions are reasonable. Also when $K > 2$.
- Naive Bayes is useful when p is very large.
- See Section 4.5 for some comparisons of logistic regression, LDA and KNN.